Architecture

PDF edition of preserved repository document | docs/architecture.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Architecture

The state contains operand entities, their multiplication relationship, decimal positions, carry, row accumulator, accumulated partial rows, result sign and goal. Each step records state_before, action, event, actual result and state_after. The original observation is retained on the trajectory. Inference, prediction, actual result and hypothetical result are distinct EvidenceKind values; execution accepts only actual results. Prediction/simulation are not implemented.

For each multiplier digit, process multiplicand digits from least significant to most significant, flush carry, accumulate the shifted partial row, then advance. A separate finish action materializes the signed answer. Goal evaluation checks completion and the independent product oracle. Verification reconstructs and replays from the observation, checking every complete transition and claimed answer. Shared replay logic catches corruption but is not an independently implemented arithmetic proof; exhaustive small cases and independent oracle tests mitigate shared implementation bugs. No memory, learning or planning exists yet.

Benchmark

PDF edition of preserved repository document | docs/benchmark.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Controlled benchmark infrastructure v1

Purpose and task

Measure accuracy, evidence-supported answers, trajectory validity, goal success, structural generalization and eventually data efficiency independently. There is no composite score. The first backend is signed decimal integer multiplication; the core state-transition implementation remains unchanged. Registry-backed dataset adapters supply sampling, identities, structure definitions, target validation and prediction assessment. Generic generation, persistence, splits, evaluation and result storage also work with non-arithmetic backends (covered by a test-only toy backend).

Schema and supervision

BenchmarkExample contains example_id, environment, problem, observation, initial_state, goal, target_answer, target_trajectory and metadata. Target trajectory stores transitions, without another full copy of observation/initial state. Metadata records seed, operand sizes, structure, split, generator version and measurable difficulty: carry_count, step_count, multiplication_steps, max_operand_digits and max_intermediate_value. carry_count counts digit operations producing nonzero carry. Optional difficulty_bounds filter those measurements, never subjective scores.

Model-facing views and export forms:

- answer_only: problem/observation; training export adds only answer.
- trajectory_supervised: also initial state/goal; training export adds trace/answer.
- experiential: observation/initial state/goal; no reference answer or trace, even in training export. Later learning must obtain feedback through environment interaction. Interactive learning is not implemented in this phase.

Evaluation always passes a deep-copied target-free view, regardless of regime. Models cannot access targets through the evaluator API. A deterministic reference executes the original environment from the observation rather than reading targets. Metadata/difficulty derived from target traces is also withheld from model views. The authoritative dataset contains targets and must never be handed directly to model training in the wrong regime. Future model integrations must enforce filesystem and procedural-knowledge access policies; Python APIs are not a security sandbox.

Generation, IID and OOD splits

TOML configs separately define counts and permitted structures for train, validation, test and optional ood_test. Explicit structures (such as 4x3) or per-split min_digits_a/max_digits_a/min_digits_b/max_digits_b ranges are supported. signed and include_zero are configurable. Each sampled target comes from the authoritative environment. Seeded split-specific RNGs and identity rejection generate numerical examples directly per split, rather than shuffling a giant dataset. Train guarantees at least one example per declared structure. IID examples use structures actually observed in train; OOD structures must be isolated, including swapped lengths.

Identity checks reject exact problem, operand-pair, example-id, and full-target-trace duplicates. Multiplication canonicalizes swapped operand pairs. Individual operand values may recur: 12×4 and 12×5 are distinct problems, not forbidden factor reuse. Shared procedural transition patterns are expected and are not trajectory leakage. The backend reconstructs every target to reject copied/tampered traces and metadata. Validation also checks split counts, membership, seed, version, difficulty and checksums. Failures are loud; records are never silently deleted. Training subsets are seeded, reproducible, nested prefixes and recorded as IDs in the manifest. Unavailable budgets are errors. Small subsets need not include every training structure.

Generation is rejection sampling, not balanced stratification. Finite small families can exhaust. In development_v1 all 45 unordered nonzero positive 1x1 pairs occur in train; IID test/validation therefore contain only 2x1 and 2x2. This distribution is reported, not a balanced per-family claim. Exhaustion raises after a bounded number

of attempts. Plan reserved per-structure quotas for final experiments if needed.

Metrics and denominators

Exact accuracy: type-strict correct final answers / all examples.

Trajectory validity: rule-consistent trajectories / supplied trajectories. Missing trajectories are excluded from this denominator; the supplied count is reported.

A valid incomplete prefix can have trajectory_valid=true but goal_success=false.

Goal success: complete verified final state AND correct claimed answer / examples.

Grounded answer rate: correct answer AND valid complete supporting trajectory bound to the same observation / examples. Correct answer-only predictions are ungrounded.

Unsupported answer rate: 1 - grounded rate. Empty denominators are null.

OOD accuracy is accuracy on ood_test; other split runs report it as null.

N*(alpha) uses only observed budget/accuracy points: minimum evaluated budget meeting alpha, or "not reached". DER = N*_baseline / N*_mindscape, or "not reached" if either threshold is unavailable. No interpolation, fabricated crossing, or deterministic reference data-efficiency claim. Alpha and observations are range-validated.

Results and reproducibility

JSONL splits and manifest.json contain exact config, generator/dataset versions, SHA-256 content hashes and nested subset IDs. Directory creation refuses overwrite.

The runner validates every split before any model call, including train contamination.

Model interface: predict(target_free_view) -> Prediction; the evaluator uses a backend, not model internals. Model exceptions and timeouts become failure records.

Each run creates a unique results/<timestamp_uuid>/ containing config.json, metrics.json, optional predictions.jsonl and summary.md. Config stores the full dataset manifest/hash, evaluation settings, Python/package versions and hardware. Metrics contain model/regime/budget/seed/split, separate rates, timings, call count and parameter count. Training time, checkpoint and hyperparameters are null for untrained references.

Prediction records include targets for post-evaluation auditing, correctness, trajectory, validity, goal status, groundedness and extensible error taxonomy.

Budget labels do not train or restrict this deterministic model's procedural knowledge. The runner is evaluation infrastructure; future training orchestration must train on the recorded subset and attach checkpoints/hyperparameters before scientific claims. Timestamps, IDs and timing vary across runs; data bytes, predictions and rates are reproducible. Results are retained locally and ignored by Git, not overwritten.

External historical references

This internal controlled benchmark is separate from historical general-model references. See historical_references.md. No external score is populated or compared. A multiplication accuracy cannot be compared with a score on another task.

Controlled numerical-claim revision

The revised study is documented in research_protocol.md and milestone_04.md. Prior singleton/oracle-assisted results are exploratory, not current evidence. Primary numerical-claim execution writes model proposals without correction; Regime C uses training-only scalar feedback and positive replay. All four conditions share capacity and the evaluator; remaining procedural/supervision/compute differences are explicit. All reported thresholds are censored; no data-efficiency advantage has been established. Use the research-specific CLI/config files for this study.

Cloud Migration

PDF edition of preserved repository document | docs/cloud_migration.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Mindscape Cloud migration checkpoint

This is a continuation of the frozen study. The local native runner was stopped through its checkpoint-boundary SIGTERM handler; all 400 locked cases are now present (A100, structured100, B100, C100). All 400 independently replayed state/tool/terminal records agree. The 1122 original CPU learning cases are byte-preserved; 798 remain. Dedicated latency has not yet run. The project remains partial and has no research-completion tag.

The private repository is akilanpl/mindscape; migration branch: mindscape-cloud-migration-2026-10-08. Starting local commit: d034a9b. The Git checkpoint includes all raw research result directories, historical immutable packages, datasets, checkpoint weights/metadata, WASI source runtime, public benchmark inputs, research logs, source/scripts/configuration/docs/tests/demo and experiment receipts. Required generated evidence is retained in versioned research_checkpoint/2026-10-08 archive parts, each at most 32 MiB, with SHA-256 per part and per restored file. Foundation model weights remain exact pinned external downloads with shipped hashes. Virtual environments, credentials, generation caches, machine auth logs, Python/OS caches and disposable compiled-runtime caches are excluded; immutable historical package bytes are preserved.

Restore into a clean destination with the existing package verifier:

```
python scripts/coding/migration_checkpoint.py restore --destination
work/migration_restore_verified
```

The unpacker verifies every archive part and the snapshot manifest, then calls restore_completion.py for complete inventory/hash validation and restoration. Restored artifacts belong under their original relative paths; source/code also exist in Git. Cloud must checkout the exact pushed checkpoint, verify it, restore the artifact archive, and route only restored results/datasets/runtime/public-input/logs into the checkout. Do not overwrite Git source files from an older restore.

Cloud continuation

Inspect actual hardware/framework/RAM/disk; never assume a GPU. Start the persistent 28800-second budget when actual Cloud continuation begins, including setup; never reset it on resume. Preserve the budget file when restoring artifacts. Bootstrap the exact pinned 1.5B backbone only; do not retrain or regenerate datasets. Validate frozen source, dataset, memory and all adapter hashes before inference.

cloud_runtime.py is an execution-only adapter around the unchanged frozen resident model/controller. CUDA uses float16 after actual tensor allocation; CPU uses the previously supported float32 reference path, explicitly recorded as an arithmetic/hardware stratum. It fixes inherited MPS-only timing labels to report the actual backend. A regression test performs real CPU neural forwards and compares greedy single/batched outputs. No scientific policy file, prompt, task, score, seed, checkpoint or training data changes.

cloud_continue.py profiles real initial requests on already exposed diagnostic tasks at batch1/4/8/16 where feasible, chooses throughput without locked-outcome tuning, and resumes only missing learning keys. It keeps the model/tokenizer resident, preserves per-case fragments durably, merges completed rows even after operational exceptions, and reserves an hour for latency/audit/report/package work. It logs CPU/CUDA

resources and actual batches. Operational failures leave incomplete cases unscored; they never receive fabricated outcomes. Dedicated latency uses 16 fresh sequential cases and actual backend labels. CPU/tokenizer/WASI/Git operations remain CPU utilities.

```
PYTHONPATH=src python scripts/coding/cloud_continue.py
```

After neural work: independently validate final locked evidence, rerun fairness, complete tests/lint/compile/build, recompute reports/plots, preserve final raw results and verify clean package restoration. If all 1920 learning cases and every mandatory release criterion pass, create mindscape-research-complete-v1 only after the final commit is pushed and independently checked. Otherwise publish a precise partial checkpoint with unrun counts, no completion tag. Original MPS evidence and historical negative studies remain intact.

Actual locked task success: A IID8/50, OOD12/50; structured IID50/50, OOD22/50; B IID45/50, OOD24/50; C IID49/50, OOD18/50. Successful replay is not successful repair. Different supervision, interaction compute, hardware and arithmetic prevent architecture-only attribution. Semantic hallucination, injected-fault recovery, zero-budget and component intervention metrics are unmeasured; supported natural self-correction is reported separately. No GPT/Gemini superiority is established.

Coding / Audit

PDF edition of preserved repository document | docs/coding/audit.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Flagship transition audit

The multiplication release mindscape-final-study-v1 remains immutable. Its 92-test implementation and frozen negative findings are historical/control evidence, not coding results. No old checkpoint or experiment is overwritten.

The generic structural Environment protocol and domain-dispatched runner can be reused. Concrete decimal State/Observation/Action records cannot represent files or edits without inappropriate fields; additive coding dataclasses are necessary. Memory storage and bounded DreamEngine already accept generic payloads. The multiplication implementation requires no changes.

The existing boolean answer evaluator cannot independently execute hidden code tests; a coding backend must bind a final repository hash to independently executed private tests. Learner-facing views must omit hidden tests, correct source and teacher patches. Visible-test feedback is permitted; hidden evaluation is terminal and never returned during the repair loop.

Current hardware: Apple M5, 10 CPUs,16GB. Local CPU PyTorch/Transformers is installed; no Docker/Podman/Ollama executable was found. Native sandbox-exec exists but the actual launch failed sandbox_apply: Operation not permitted. Host Python execution of generated code is therefore forbidden. The planned fallback is WASI Python through Wasmtime, preopening only a disposable task directory and curated Python standard library, with no network or unrelated host filesystem capability. Probe file/network/process denial, timeout and memory limits before any benchmark code executes. Fail closed if the runtime cannot enforce them.

Two Apache2.0 coding backbones (Qwen2.5-Coder0.5B and1.5B Instruct) are candidates. Actual pilot capability and CPU latency, not nominal size, will select the practical backbone. Model-only generation, structured generation and real bounded repair loops must remain distinct. Domain-data adaptation must be labeled precisely; retrieval/sample budgets are not gradient-training budgets. Unequal interaction/supervision/compute remains an explicit comparison limit.

Coding / Final Architecture

PDF edition of preserved repository document | docs/coding/final_architecture.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Coding research architecture

The historical integer multiplication environment and release remain intact. Coding is a separate adapter over the existing Goal, Entity, Relation, benchmark and Prediction interfaces. Its task/state/action/event/result/transition/trajectory records represent actual repositories and tool calls, rather than arithmetic-specific operations.

A learner receives the broken repository, specification and visible tests. Private test inputs and expected values, reference sources and expert patches remain in dataset/evaluator records outside its observation. Training-only expert trajectories are actual executed inspections, visible failures, accepted repairs and visible verification; hypothetical plans are excluded from verified experience memory. The benchmark wrapper constructs a learner task without private cases and delegates terminal assessment to an independent backend.

The thirteen typed coding tools inspect repositories, files, symbols and visible tests, search, edit or apply a unique patch, run one/all visible tests, inspect errors, revert, compare versions and finish. Repository paths reject absolute/traversal names. Real test execution uses a pinned CPython 3.14.7 WASI guest, with capability-limited filesystem mounts, resource fuel, bounded output, memory and wall time. No model-generated Python runs in the host interpreter. Host-side expected-value comparison requires matching value and type, not merely process exit zero.

The primary backbone is Qwen2.5-Coder-1.5B-Instruct revision 2e1fd397ee46e1388853d2af2c993145b0f1098a. It runs locally in float32 on CPU. Rank-eight q_proj/v_proj LoRA trains 1,089,536 parameters. Four conditions share this backbone and nested unique training-task budgets: plain repair, structured repair, persistent typed-action policy and actual-experience replay. Supervision, input tokens, runtime feedback and interaction budgets differ; this is not an architecture-only causal comparison. Replay updates policy weights using accepted repair actions and real past feedback. It is offline expert replay, not reinforcement learning or self-play. Inspection/testing/stopping actions rely on the pretrained model: accepted repair is the only trained action type.

Working memory resets per episode. Verified episodic memory contains only executed training examples; specification-word retrieval and concept records are bounded. Evaluation repairs never become verified training experience. The relation graph records files, symbols and calls. The planner compares two bounded branches using static preconditions/syntax. Predicted states are marked hypothetical, and predicted test success is null. It is not a learned world model; actual execution alone can establish success.

The secondary SQL environment reuses Goal/Entity/Relation and Model/Prediction records with typed schema/table inspection, query editing, execution and stopping. Its in-memory SQLite authorizer permits SELECT capabilities, with an execution deadline, output limits and no extension/file access. Private database rows and expected queries are terminal-only. The small SQL study tests interface portability without SQL fine-tuning.

Coding / Final Experimental Protocol

PDF edition of preserved repository document | docs/coding/final_experimental_protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Frozen experimental protocol

Preserve original protocols, results and multiplication negative findings. Recover useful jobs and checkpoints; distinguish teacher collection, retrieval adaptation and gradient training. The recovery inventory records deliberately deferred retrieval inference and its retained 2,400-example accepted prefix.

The frozen generator dataset contains 5,000 training, 20 validation, 20 IID test and 20 structural OOD tasks. Ten training categories cover simple Python bugs; four OOD categories hold out function interaction, cross-file repair, compound bugs and unfamiliar filtering structure. Every correct/buggy pair is actually executed. IID shares semantic templates with training intentionally; foundation-model contamination remains unknown. Learning pools have prior diagnostic exposure, disclosed explicitly.

The independent final lockbox contains 100 fresh tasks: 50 IID and 50 OOD, seed 13001, excluding every prior training/validation/evaluation seed. Its SHA-256 is 392ef68a412f8df49b93af60ee24318a6746a78aaeee30acdc0d6f72e92bc8a5. The final configuration is fixed at training budget 100, seed 11, three edits/eight actions, before lockbox results. Evaluate each condition once; no lockbox-driven repair, tuning or selection. Deterministic resumability reuses completed episodes and does not resample unsuccessful tasks.

The gradient comparison measures budgets 10, 25, 50 and 100 with three independent adapter initialization seeds 11,23,37. Plain-patch training additionally retains seeds 53,71. Nested checkpoints follow identical ordered task prefixes and one training epoch; examples are unique tasks, not optimizer steps, tokens or retrieved demonstrations. A zero-domain-training control supplies N=0. Requested larger gradient budgets are unexecuted and cannot be inferred from collecting 5,000 trajectories. The practical compute limit is documented using actual observed training time/memory. This reduced range right-censors unresolved thresholds.

Accuracy uncertainty averages observed seed outcomes within each task and bootstraps paired task clusters, 2,000 draws, fixed analysis seed17. Report observed per-seed means and sample SD separately. These task-cluster intervals do not establish a population-wide training-seed interval. N* is the minimum observed budget reaching 80/90/95 percent, without interpolation. Unreached thresholds are right censored; DER is undefined for missing thresholds or zero denominator. Normalized trapezoidal AULC integrates accuracy over log positive budgets; N=0 is reported separately. Paired deltas compare identical task keys. Public pass@k is never inferred from interacting edits.

Actual post-termination snapshot verification supplies interactive repair-proposal success@1/3/5 in the recovery diagnostic. Invalid unparseable proposals count as failed attempts; inspection/testing actions are excluded. First-applied-patch success in the final study is separately labeled and is not single-call capability. Syntax, incorrect-return, regression and nontermination faults are executed visibly. No private verdict is fed

back during policy actions. The final study allows at most three edits; success@5 is unavailable there. Invalid parsed actions, invalid executed transitions, terminal goal and grounded success are distinct quantities.

Inference ablations remove state, relations, memory, explicit goal, exemplar trajectory, environmental execution feedback or hypothetical planning while keeping trained weights fixed. They do not remove latent knowledge learned during training. Robustness perturbations rename functions/variables, reorder files/tests, add distractors, relocate packages and reformat source. Each perturbation's oracle and defect are actually checked; unchanged/inapplicable variants are omitted. Small diagnostic samples do not substantiate universal robustness.

Latency runs disable generation caching and run after other model experiments. Report per-episode inclusive stage times, fresh token-generation TTFT, wall time and model/tool/token counts. Stage timings are nested, so cannot be summed as exclusive components. Fourteen cases per condition make P99 exploratory. Ordinary OS activity is uncontrolled. Cached research episodes remain valid accuracy records but cannot support fresh performance claims.

Public HumanEval uses all164 official pinned tasks, one greedy completion, max512 new tokens, full-module instruction prompts and actual WASI execution. It has no adaptation, few-shot examples or feedback. Report pass@1 only. Preserve the earlier 0.5B result; compare the selected1.5B model under the same local protocol. This prompt/runtime differs from historical API reports, whose exact sampling settings may be undisclosed. No general GPT/Gemini superiority claim follows.

Robustness/ablation runs also disable generation caching, allowing measured dream/memory timing deltas. A supplemental no-trajectory-supervision diagnostic switches to the patch-only adapter at the same100-task budget/seed and keeps the typed action loop. Training input/label changes make this a combined intervention rather than isolated removal. The explicit-goal ablation removes policy goal text; terminal private assessment remains necessary to measure success. A “no terminal goal evaluator” result would be unmeasurable and is not claimed.

TTFT measures the first decoder token after generation starts, excluding model load and preceding prompt tokenization. Peak RSS is the host process high-water mark; fresh latency shares one process across sequential configurations, so its per-row RSS is not an isolated per-task peak. WASI guest limits are separately recorded. Test-suite executions count actual VM calls, including terminal assessment, rather than claiming every expected case ran when import/compile fails.

Coding / Historical Gpt Gemini Comparison

PDF edition of preserved repository document | docs/coding/historical_gpt_gemini_comparison.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Historical HumanEval references

These are historical report values, not newly rerun API models. Our scores use the same benchmark task family but a different instruction prompt and CPython3.14.7 WASI runtime. Neither sampling nor contamination control is matched sufficiently to establish general superiority.

Historical model	Reported HumanEval score	Reported prompting	Primary source
GPT-3.5	48.1%	0-shot	[GPT-4 technical report, Table2](https://cdn.openai.com/papers/gpt-4.pdf)
GPT-4	67.0%	0-shot	[GPT-4 technical report, Table2](https://cdn.openai.com/papers/gpt-4.pdf)
Gemini1.0 Pro	67.7%	0-shot, instruction-tuned	[Gemini report v1, Table2 p7](https://arxiv.org/pdf/2312.11805v1)
Gemini1.0 Ultra	74.4%	0-shot, instruction-tuned	[Gemini report v1, Table2 p7](https://arxiv.org/pdf/2312.11805v1)

The GPT report was initially published March2023; exact API checkpoint identifiers and complete HumanEval decoding/sample settings are not specified in that table. The Gemini report v1 is December2023, with table API comparisons collected November2023. Its instruction-tuned flag does not supply a complete replicable prompt/decoding protocol. No temperature, number of samples, parameter count or hidden model revision is inferred from omitted information. Values are treated as published functional-correctness percentages; our measured statistic is one-completion pass@1, and raw numerical differences are descriptive only.

The pinned Gemini v1 PDF SHA-256 is f3c5b175d26159b92891b32380515ec1036f4ffe2862e19d7277fd165e0d9663. A current hosted report has later revisions, so v1 is used for historical attribution. The official HumanEval repository (https://github.com/openai/human-eval) supplies the164 tasks. Pretraining contamination of our open models is unknown. Local split disjointness cannot resolve this limitation.

The earlier0.5B open-model run remains94/164,57.317073% under its recorded local greedy512-token protocol. The final measured1.5B run is reported separately in the machine scorecard after completion. Claims such as “beats GPT-4,” “contamination-free” or “general coding perfection” are not supported by this comparison.

Coding / Mps Acceleration

PDF edition of preserved repository document | docs/coding/mps_acceleration.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Native MPS execution

The central device manager defaults to MPS and validates allocation before loading weights. Final neural evaluation refuses CPU placement and PYTORCH_ENABLE_MPS_FALLBACK=1. Explicit CPU reproduction is available for historical experiments; it is never an automatic substitute for the accelerated benchmark.

One resident backbone and tokenizer serve independent episodes through a dynamic request queue. Workers own their environment and memory; only the coordinator performs neural computation. Each episode's next request follows its actual previous tool result. Padding is left-sided to multiples of 64; contexts are not truncated. Model parameters, buffers, input tensors and LoRA computation reside on MPS. The train scripts explicitly place token tensors and model gradients on the selected device. Adam optimizer state follows parameters; scalar accounting and serialization remain host utilities.

The device audit found only explicitly placed training tensor constructors and the device-allocation probe. Adapter artifact validation opens CPU safetensor metadata without a forward pass. The retained CPU checkpoint backup is used to restore exact original values across precision trials, then released. Tokenization, decoding after inference, repository access, WASI execution, hidden tests, result serialization and Git remain CPU work.

The native runner tests float16/bfloat16 support and batches 1, 4, 8 and 16 within a ten-minute profiling cap. It compares fresh MPS episodes with preserved uncached CPU episodes from the identical checkpoint and prompts. Precision, padding and batching can affect greedy output; all observed correctness differences must be reported. Historical float32 learning cells remain separate from reduced-precision MPS cells. The speed comparison combines hardware, precision and batching; it is not an isolated GPU causal estimate.

Completed episodes are fsynced immediately and skipped on resume. The lockbox configuration is written before any locked outcomes. The original two-hour deadline is fixed, not reset by a process restart. Missing work remains explicitly not run.

The execution environment cannot allocate MPS despite native Terminal availability. The user launches the native command; tooling monitors workspace outputs and conducts CPU/WASI evidence checks. Native device placement and allocator measurements establish inference use; availability alone does not.

References: PyTorch MPS (https://docs.pytorch.org/docs/main/notes/mps.html), MPS fallback environment variable (https://docs.pytorch.org/docs/stable/mps_environment_variables.html).

Coding / Protocol

PDF edition of preserved repository document | docs/coding/protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Coding protocol v1

The flagship implements bounded Python repair with actual edits to disposable repositories and actual function execution in capability-isolated CPython WASI. The host compares private expected values. Model-visible observations contain visible tests only. Hidden evaluation occurs after the policy stops. Goal verification cannot be satisfied by a model's success declaration.

The selected model is Qwen2.5-Coder-1.5B-Instruct, revision 2e1fd397ee46e1388853d2af2c993145b0f1098a, Apache 2.0. A locked ten-task validation pilot produced 9/10 actual repairs, versus 7/10 for the 0.5B model. This small selection pilot is not a general capability estimate. Full model weights remain outside Git in the local cache.

Four policies share this backbone: model-only source repair, structured state repair, trajectory-example repair, and bounded experiential repair. The last policy sees actual visible-test feedback and has up to three sequential attempts. Its success@3 is an interactive recovery measure, not HumanEval pass@3. Its memory uses executed teacher examples. The first two controls do not acquire memory as the sample budget changes, so their frozen results are reused transparently rather than counted as new independent seeds.

The first matrix measures retrieval adaptation at nine sample budgets (10 through 5000) and five independently shuffled training-memory seeds. These are retrieval budgets, not gradient-training budgets; gradient DER is not identifiable from this matrix. Teacher patches and trajectories constitute more supervision than the controls receive. Extra tool execution and retries constitute more inference compute. Any difference is conditional on those information and compute differences.

Task families are intentionally simple synthetic Python functions. IID splits share structural families but distinct names, parameters, and repository hashes. OOD families are withheld from training: function composition, cross-file dependency, compound repair, and filtering structure. These four families do not represent broad repository engineering. Exact-hash nonoverlap cannot establish semantic independence for shared IID templates. Public benchmark pretraining contamination is unknown.

WASI preopens only curated standard-library files and the current disposable task repository, both read-only to generated code. Host tools apply validated edits. Runtime probes deny unrelated file access, sockets, and subprocess execution. Fuel, memory, CPU, output-file size, and host wall deadlines bound execution. Native macOS sandbox creation failed with Operation not permitted; Docker and Podman are absent. Generated code never executes in host Python.

The pinned official HumanEval revision is 6d43fb980f9fee3c892a914eda09951f772ad10d (164 tasks). Independent greedy completions receive no hidden-test feedback. CPython WASI differs from the original host runtime; report incompatibilities as failures and publish the runtime. With one completion per task, only pass@1 is available. No pass@k estimate is produced for k greater than one.

train_lora.py provides separate real LoRA gradient adaptation (rank 8, query/value projections, masked prompt loss, AdamW, one epoch). Its resulting records and checkpoints must not be conflated with the retrieval matrix. No gradient threshold claim is permitted without evaluating checkpoints at matched training budgets.

Exact prompt/model/revision/token-limit repeats under greedy sampling may reuse an on-disk completion cache. Cached calls are recorded with zero new model calls/tokens and zero generation time, plus the original measured timing. Actual repository execution is repeated for each evaluated policy. Cache hits are not additional independent model samples; latency summaries must distinguish fresh calls from reuse. Changing the model revision or any prompt byte changes the cache key.

The earlier 1000-task/three-seed configuration was a preflight only: no held-out model evaluations were run. Its locked validation/test/OOD tasks are retained during expansion to 5000 training tasks; all held-out seeds are excluded from the additional training pool. Executed prefix teacher records are reused with their actual provenance.

The independent LoRA diagnostic exposed a source-envelope mismatch: correct-looking JSON repository maps were rejected by the initial path/content-only parser. A general normalization now accepts a single repository-map edit, a typed edit, path/content, or a source block whose target is unambiguously identified by the public entry point. Source text is never repaired by the parser. Ambiguous and escaping paths remain rejected. The original held-out sets are therefore diagnostic, retained in full; a fresh primary set (seed 9001, 20 IID/20 structural OOD tasks) was frozen before primary comparisons. Training remains the same 5000-task pool, and fresh held-out seeds exclude all training and prior evaluation seeds. This disclosed protocol revision prevents diagnostic exposure from masquerading as untouched testing.

Full typed expert tuples are archived separately from the compact model-facing examples. The accepted preflight prefix is reexecuted to collect genuine state/action/event/result/next-state records; archived results are never synthesized from labels. A teacher repair that fails actual visible execution stops collection. These provenance additions do not change model prompts or proposed source edits.

Completion adds a bounded two-branch static planner. Each proposed action and alternative produces a hypothetical state, a hypothetical follow-up test action, and a heuristic precondition/syntax score. This is not a learned world model and never predicts authoritative test success. Planning can reject an invalid predicted edit in favor of a real observation action; the no-dream ablation measures its actual effect. Stage timers retain full durations in transcripts; irrelevant timing metadata is excluded from model prompts for deterministic prompt reuse. Previously measured adapter results and old protocol records are retained as their original controller versions.

Frozen completion comparison

The preceding sections preserve the historical retrieval/diagnostic protocol. The completion comparison uses pinned Qwen2.5-Coder-1.5B LoRA checkpoints, gradient budgets 10/25/50/100, learning seeds 11/23/37 and 20 IID plus 20 OOD tasks per cell. The independent lockbox uses seed13001, 50 IID plus 50 structural OOD tasks, checkpoint budget100 and training seed11 in each of A/structured/B/C. A and structured produce one patch; B/C retain up to eight proposals and three accepted edits. The exact frozen source/checkpoint/task hashes are in emergency_mps_v1/locked_protocol.json.

The operational continuation uses frozen batch16/float16/MPS, keeps model/tokenizer resident and disables generation caching. Original CPU float32 learning evidence remains unchanged and separately identified. The original emergency deadline is historical metadata; the later completion request removes that operational cutoff only. No new scoring, held-out tuning, hidden feedback or altered training data is introduced. Proposal

rejection is separate from accepted-transition validity; actual repair success is independently assessed at terminal execution.

Coding / Recovery Inventory

PDF edition of preserved repository document | docs/coding/recovery_inventory.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Completion recovery inventory

Recovered on main at 4689e6d; multiplication commits/results remain intact. Four useful jobs survived: full expert prefix tuple collection, 5000-example actual teacher collection/retrieval sweep, five-seed LoRA prefix training (10/25/50/100), and the initial primary adapter evaluation. None was terminated. The adapter evaluation completed and is reused; the other jobs continue from their existing checkpoints.

Historical 0.5B HumanEval: 94/164 = 57.317073% under greedy full-module instruction generation, CPython 3.14.7 WASI, 512 output tokens, one sample/task. It is preserved, with its noncanonical runtime documented. All 5060 generated correct/buggy pairs were actually validated (40480 function invocations); model-generated programs are never batched or run in host Python.

Completed 10-example adapter evaluation: plain baseline 18/20 IID, 10/20 OOD; structured baseline 20/20 IID, 9/20 OOD. These 40 cases are the primary learning-curve pool, not the later one-shot bounded-perfection set. The completion protocol calls for a separate 100-task lockbox, frozen before final model selection.

The existing large sweep is retrieval adaptation, not gradient data efficiency. The existing gradient checkpoints are patch-supervised; they cannot alone substantiate trajectory learning or experiential replay. Completion adds matched structured/action training and actual accepted-experience replay, with supervision differences audited explicitly. No DER or architecture advantage is inferred from the preliminary numbers.

Completion lockbox: 100 fresh tasks (50 IID/50 OOD), generation seed 13001, all prior training/evaluation seeds excluded. Preselected final budget100/seed11; no score-based configuration selection on this set. The 40-case learning-curve pool remains distinct and its prior diagnostic exposure is disclosed.

The surviving teacher/retrieval job was checkpointed deliberately before any retrieval inference to prioritize the new gradient objective. Its accepted teacher prefix is reused in a collection-only continuation; original protocol/checkpoints remain intact. This avoids running the old retrieval-only controller for hours as if it answered gradient data efficiency. Its supplemental inference sweep can run separately with the revised frozen controller. No completed evaluation was discarded.

Coding / Reproducibility

PDF edition of preserved repository document | docs/coding/reproducibility.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Reproduction and immutable evidence

Use Python3.12 and the measured versions in experiments/coding_completion_v2/requirements-measured.txt. Preserved training/evaluation uses CPU float32; the authorized continuation uses actual MPS float16 with independent requests batched up to 16. No paid services are used. Initial model/runtime downloads require network access; evaluated programs have no network dependency or capability. The bootstrap script pins HF revisions and validates the WASI archive checksum.

Run from the repository root, in a fresh checkout/output directory. Existing frozen datasets and final snapshots intentionally refuse overwrite. Preserve existing results before reproducing; never delete historical artifacts. The research scripts use versioned root paths, and exact generation-cache keys include model revision and adapter-weight SHA. Cache hits preserve actual prior responses but have zero newly consumed model calls/tokens/time. Fresh latency disables this cache.

```
python3.12 -m venv work/final-venv
work/final-venv/bin/python -m pip install -r
experiments/coding_completion_v2/requirements-measured.txt
export PYTHONPATH=src
work/final-venv/bin/python scripts/coding/bootstrap.py
work/final-venv/bin/python scripts/coding/reconstruct_datasets.py
work/final-venv/bin/python scripts/coding/freeze_dataset.py
work/final-venv/bin/python scripts/coding/validate_dataset.py
work/final-venv/bin/python scripts/coding/freeze_lockbox.py
```

The model argument is the downloaded snapshot directory with the exact1.5B revision. Set a task-specific shell variable to that directory (do not overwrite HOME or CODEX_HOME). The following sequence makes the shared dependency artifacts concrete:

```
mindscape_model=work/coding/hf/models--Qwen--
Qwen2.5-Coder-1.5B-Instruct/snapshots/2e1fd397ee46e1388853d2af2c993145b0f1098a
work/final-venv/bin/python scripts/coding/study.py --model "$mindscape_model" --train 5000
 --test 20 --budgets 10,25,50,100,250,500,1000,2500,5000 --seeds 11,23,37,53,71 --output
results/coding/completion_collection_v1 --locked-from
results/coding/final_dataset_v1/dataset.json --collect-only
```

On the original completion run, accepted records were recovered from a prior collector, not regenerated. For a fresh reproduction, copy the accepted compact training-only memory to results/coding/final_retrieval_v1/teacher_memory.json because the gradient comparison reads that historical-compatible path; the full tuple archive remains results/coding/teacher_traces_v1. This is data routing, not result fabrication. Actual collection produces and verifies these files before training. No held-out correct source or private feedback belongs in learner inputs.

```
work/final-venv/bin/python scripts/coding/gradient_pipeline.py --model "$mindscape_model"
work/final-venv/bin/python scripts/coding/completion_gradient.py --model
"$mindscape_model"
work/final-venv/bin/python scripts/coding/lockbox_eval.py --model "$mindscape_model"
work/final-venv/bin/python scripts/coding/latency_probe.py --model "$mindscape_model"
work/final-venv/bin/python scripts/coding/verify_final_traces.py
work/final-venv/bin/python scripts/coding/completion_analysis.py
```

```
work/final-venv/bin/python scripts/coding/build_demo.py
work/final-venv/bin/python -m pytest -q
work/final-venv/bin/python -m ruff check src/mindscape/coding src/mindscape/sql
scripts/coding tests/test_coding* tests/test_sql_vertical.py tests/test_lora_checkpoint.py
```

Public HumanEval is separately downloaded at its recorded official revision with download_humaneval.py and run using humaneval.py --model SNAPSHOT --output NEW_VERSIONED_ROOT. Use the exact recorded greedy512-token, one-completion, no-feedback configuration. Never tune on the final lockbox or public results. fairness_audit.py and training_token_audit.py provide automated local audits and exact token recounting; the final report distinguishes these from unknown foundation-model contamination.

The recorded demo is demo/coding/index.html, generated exclusively from completed final episodes. Open it directly or serve locally with python -m http.server --bind 127.0.0.1 --directory demo/coding 8765. It shows actual typed actions and visible execution, omits private expected values and raw model reasoning, and labels playback. Its timing is not the fresh latency benchmark.

freeze_emergency.py applies the strict 400-lockbox/1920-learning/dedicated-latency publication gate and copies completed data, actual tuples, adapters, raw results, protocols, scripts, docs and demo into results/final/coding_research_v2, emits SHA-256 for every frozen file and refuses overwrite. Foundation weights remain in the pinned local HF cache; their public revision identifiers and bootstrap commands are recorded. The historical mindscape-final-study-v1 release remains intact. Different hardware/library kernels can produce numerical or generation differences; bitwise cross-platform equivalence is not claimed.

The full historical source tree has201 pre-existing Ruff style findings, independently confirmed against commit3e2538f. This completion introduces none; the scoped coding/SQL checks pass. The complete functional test suite remains the acceptance gate.

Restore the immutable package

To verify every frozen SHA-256 and reconstruct the original repository layout in a new empty directory:

```
work/final-venv/bin/python scripts/coding/restore_completion.py --snapshot
results/final/coding_research_v2 --destination work/restored_research_v2
```

The restore command refuses a nonempty destination. Shared configurations, scripts, documentation, experiments, tests, source code and actual result/checkpoint files are included. It does not create or reset Git history. Foundation weights remain a separately pinned download. Use the release tag for the complete historical Git repository.

Exact completed-study evidence

The authoritative analysis is scripts/coding/emergency_report.py, with the original scientific configuration in results/coding/emergency_mps_v1/locked_protocol.json. complete_research.py resumes only missing frozen keys on native MPS; it does not reprofile or tune on locked outcomes. finalize_research.py --wait performs independent replay, local fairness audits, complete tests, report generation, strict publication gating, freezing and clean restoration. Run it only in the study checkout; it may create the completion tag only after all gates pass.

The commands above include historical data/training entry points, not an instruction to rerun a saved completed study. Restore the final package and recompute metrics directly for exact evidence reproduction. A

newly generated study must use a fresh directory, record its actual device/precision, and retain separate CPU/MPS strata; cross-kernel bitwise generation identity is not promised. The original 1122 CPU rows are byte-preserved, while the remaining cases use the measured MPS configuration.

Zero-budget, injected-recovery, SQL model evaluation, and component-ablation entry points exist but were not executed in this completion scope. They are optional new experiments and cannot supply claims for the saved release. The completion scope is exactly four conditions × 100 locked tasks, four conditions × three seeds × four budgets × 40 learning tasks, and 16 dedicated sequential latency cases. Sequential latency uses actual batch1; batch16 evaluation throughput is reported separately.

Decisions / 001-Foundation

PDF edition of preserved repository document | docs/decisions/001-foundation.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Foundation choices

Use standard-library frozen dataclasses and TOML (Python 3.11 tomllib) to keep local/offline execution dependency-free. Domain implementation is centralized in environment.py rather than duplicated across state/actions/transitions modules. Use unittest tests that are also discoverable by pytest. Empty package and directory placeholders reserve future layers, not completed capabilities.

Experimental Protocol

PDF edition of preserved repository document | docs/experimental_protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Experimental protocol

Freeze datasets and holdout rules before fitting. Record random seed, complete configuration, model identifier/parameter count, dataset version/hash/size, regime, split, hyperparameters, checkpoint, software versions, hardware and runtime. Create unique result directories; never overwrite. Save config, metrics, predictions and summary. Retain failed runs. Use 3-5 seeds where practical, mean and standard deviation and appropriate confidence intervals. Raw graph data must remain available. Ablations must remove operational components rather than only toggling labels. Historical references require exact protocol compatibility; no comparison across unrelated benchmark tasks is valid.

Phase 2 operational rules

Freeze dataset_version, generator_version, manifest content hash and split structures before fitting. The current development dataset tests infrastructure only and may inform infrastructure debugging; do not treat it as an untouched final research test. Allocate new versioned final datasets before neural experiments. Any subsequent validation-informed design or split revision must be recorded.

Train only from a recorded nested subset and regime-specific training view. Regime C receives concepts/limited examples/environment feedback rather than full trace labels; this phase provides its data view, not a learning algorithm. Record procedural knowledge access as an experimental condition. A reference algorithm is not a fair learned baseline and its perfect arithmetic score is not evidence of data efficiency.

Evaluate baseline, structured baseline and Mindscape on identical IID/OOD splits, backbone/parameter count, decoding, epochs and compute where practical. Repeat 3-5 seeds for final studies. Keep accuracy, groundedness, trajectory validity, goal success,

OOD accuracy, runtime and data efficiency separate. N* uses observed sizes only and unreached thresholds remain censored. Retain failures and every prediction file. Run directories record null for unmeasured training/checkpoint fields; future training must populate them. Final test sets must not drive tuning.

First learned development protocol

Before evaluation, record fairness.json: same dataset hash/subset IDs, train/validation and IID/OOD sets, same hidden backbone, seeds, optimizer steps, batch size, learning rate, evaluator and validation-loss selection. Head sizes and supervision count differ. Policy-only access to exact arithmetic transitions and singleton action masks are major confounds. Include unmasked policy runs and raw action validation diagnostics; never infer architecture superiority from the forced masked execution.

The fixed schedule uses budgets 50/100/250; seeds 0/1/2 at 50, seed 0 at larger budgets. These are development results on development_v1, not final untouched thesis results. The protocol and implementation are committed before evaluation. No model/prompt or test-set change is made in response to held-out scores. Match or explicitly vary procedural knowledge/tool access in later experiments. Do not compute DER from the masked algorithm's guaranteed accuracy. Historical reference scores remain unpopulated.

Controlled numerical-claim revision

The revised study is documented in research_protocol.md and milestone_04.md. Prior singleton/oracle-assisted results are exploratory, not current evidence. Primary numerical-claim execution writes model proposals without correction; Regime C uses training-only scalar feedback and positive replay. All four conditions share capacity and the evaluator; remaining procedural/supervision/compute differences are explicit. All reported thresholds are censored; no data-efficiency advantage has been established. Use the research-specific CLI/config files for this study.

Final Architecture

PDF edition of preserved repository document | docs/final_architecture.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final architecture

The reusable core stores entities, relations, observations, goals, states, actions, events, results and trajectories as typed immutable records. The generic structural Environment protocol defines reset/state/goal/legal-actions/step/goal-reached/trajectory, parameterized by observation/state/action/transition/trajectory types; both current environments satisfy its runtime structural check. Dataset backends implement generation, canonical identity, strict verification and evaluation annotation; the shared runner does not implement arithmetic. Current State includes decimal-specific fields, so arbitrary domains would require an adapter or a generalized payload schema. This is an honest boundary of reuse.

Multiplication has two retained environments: historical exact digit mechanics and controlled numerical claims. In the latter, the environment exposes a hand-designed long-multiplication cursor and accepts all 100 numerical claims. It writes the learner's proposed units/carry even when numerically wrong. Only the private reference generator, training-feedback boundary and post-prediction verifier compute the expected claim. Inference never gets a corrected digit or numerical verifier feedback.

Model-visible state uses operand digits, cursor, lengths, current accumulated digit, carry, phase and static goal/relation flags. There is no product or future correct state. The final adapter passes indexed numerical features through a frozen SmolLM2 transformer and concatenates a categorical encoding of those same features. A two-layer learned head predicts decimal answer digits for conventional A/structured conditions or units/carry for procedural B/C. All share one checkpoint revision, backbone and head capacity; training masks only inactive policy heads. This is frozen-feature task adaptation, not full language-model fine-tuning. The language model's indexed inputs limit how much natural-language pretraining capability is being tested.

Working memory follows actual transitions. Episodic SQLite memory stores training-only own attempted experiences and binary feedback; C reuses only accepted own actions. Concept memory stores descriptive text and does not answer arithmetic. Its presence alone is not a measured benefit. The matched-information conventional local policy uses B's exact predictor/state labels and real execution but bypasses the memory wrapper.

The bounded dream engine creates tagged hypothetical states using proposed-state dynamics, never real environment evidence. Its arithmetic scoring is model confidence, not a correctness oracle. The generic integer-position goal demo separately demonstrates goal-directed bounded rollout; it is not a learned second-domain experiment. Final decisions tag real/hypothetical results, record100 candidates and proposals, and receive correct-action/transition judgments only after prediction.

Final artifacts include strict dataset manifests, head NPZ checkpoints, backbone revision metadata, configs, full predictions, structured evidence, failure groups, resource counters and statistics. The frozen model remains local and offline after its initial download. The old NumPy MLP and all failed studies are preserved.

The running final study loaded its original annotation class at protocol freeze. Real decisions have inline posthoc correct-action judgments. No-interaction decisions additionally receive immutable hypothetical_judgments.json sidecars after evaluation; raw predictions are retained. The current backend also supports those posthoc hypothetical annotations for fresh reproductions. Neither route supplies oracle values to inference or promotes hypothetical agreement to real trajectory validity.

Final Audit

PDF edition of preserved repository document | docs/final_audit.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Audit before final model changes

The phase-4 artifacts and all historical code remain intact. The audit used source inspection, the existing numerical-gradient regression, dataset validation and executed diagnostics in experiments/final_audit/.

Component	Finding
Input/numeric representation	Answer baseline uses ordinal decimal digits; policy uses four one-hot local digits. This information/representation difference matters. Neither receives the final answer.
Labels/output/loss	Nine independent categorical heads; policy uses only two numerical heads while seven constant labels consume 7/9 of the loss. Answer labels contain many padding zeros. High per-digit accuracy is misleading for exact arithmetic.
State/targets	Strict typed serialization; reference targets replay authoritative claims. Own wrong claims are executed unchanged during inference. No backfilled trace.
Optimizer/normalization/batching	Adam implementation and gradients have regression coverage. Fixed 600 steps, replacement batches, bounded features. Six thousand steps improve tiny-set fit; this does not isolate one causal explanation.
Splits/OOD	Canonical operand identity prevents swapped-pair leakage. Structures are held out. All 45 nonzero canonical 1x1 pairs are exhausted in training, so the IID validation/test contain no 1x1 examples.
Seeds/checkpoints/decoding	Seeded initialization and minibatches; saved arrays checked for shape/finite values. Independent units/carry decoding makes exact claims harder. A skip score uses the sign head, a semantically artificial coupling.
Metrics/execution	Exact answer, entire verified trajectory and goal are separate. Wrong claims can cancel to a correct answer. Baselines without trajectories cannot satisfy evidence-based groundedness.
Candidates	101 open candidates or 100 syntactically legal numerical values. No singleton correctness mask.
Supervision/information	B gets all transition labels; C gets bounded binary feedback on own proposals. Policy state exposes a hand-designed long-multiplication cursor. This is domain structure, not evidence of discovering a procedure.

Component	Finding
Costs/class imbalance	C's 64 attempts often cannot finish an episode. Accepted replay is sparse and biased toward early states. Constant heads/padding dominate labels. B's multiple labels per problem are not comparable to one answer label without qualification. Dream increases model calls.

Executed same-set diagnostics at seed 0 and 6000 updates: 10-example answer and policy accuracy both 100%; 45-example 1x1 answer accuracy 46.67%, policy accuracy 100%. These are memorization tests, not benchmark results. Further carry/within-structure diagnostics are recorded separately. The pipeline can learn and execute; the phase-4 failure is not explained by a universally broken label/gradient/decoder pipeline. Capacity, representation, optimization budget, feedback sparsity and distribution coverage remain plausible interacting causes.

Final interventions must preserve this negative study. Remove dummy policy losses and artificial skip/sign coupling in the new adapter, use an actual pretrained local backbone and a richer learned head, retain explicit resource and information differences, and freeze the new protocol before inspecting its held-out results. No intervention supplies computed arithmetic as a feature.

The additional seed-0 n=100 trajectory diagnostic memorized its training set at 100%; on validation it reached 17% exact answers, 66.46% local claims and 81.43% carry labels. Thus local errors compound during execution. These measurements use validation only and do not select final-study hyperparameters.

The final representation also passed executed sanity checks: answer-only/structured/B memorized their n=10 training subsets at 100%; C achieved0% with incomplete accepted experience. All45 canonical1x1 problems and a mixed100-example set were memorized at 100% by final answer/B heads with 1200 updates. On the new validation set the separately trained mixed100 diagnostic reached3% answer-only and18% B exact accuracy; B local-claim accuracy62.47%, carry accuracy82.60%. C's missing successful sequences are a feedback-coverage failure, not evidence that the supervised pipeline cannot memorize. These diagnostics were recorded after protocol freeze and did not change hyperparameters.

Final Claims

PDF edition of preserved repository document | docs/final_claims.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Claims and evidence

Claim	Status
high bounded-vertical capability	NOT SUPPORTED
OOD improvement	SUPPORTED
data efficiency	INCONCLUSIVE
grounded execution	SUPPORTED
trajectory validity	SUPPORTED
goal success	NOT SUPPORTED
recovery	INCONCLUSIVE
external benchmark competitiveness	INCONCLUSIVE
memory benefit	INCONCLUSIVE
dream benefit	INCONCLUSIVE
GPT reference exceedance	NOT SUPPORTED
Gemini reference exceedance	NOT SUPPORTED

Capability/goal claims use a descriptive 90% threshold (reporting criterion, not a preregistered hypothesis test) on both locked splits. OOD improvement requires a positive task-paired 95% CI against A. Grounded execution requires all locked independent replay evidence to match. Grounding is established only for saved re-executed episodes. No architecture-only causal claim is supported.

Final Experimental Protocol

PDF edition of preserved repository document | docs/final_experimental_protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Frozen final protocol

The historical 8722-parameter MLP is retained. Final primary conditions use the same frozen Apache-2.0 SmolLM2-135M-Instruct transformer and the same two-layer task-head architecture, initialization seed and nested training problem IDs. Pretrained weights are not fine-tuned: this is a frozen-feature adaptation study, not an instruction-tuned arithmetic model or a claim that a language model is intrinsically stronger at this task. Its actual results will determine quality.

Canonical model inputs are numerical operands for answer-only, operands with goal/length markers for structured, and current cursor/local operand digits/partial digit/carry/position/phase for procedural B/C. No product, expected claim, reference future state or verifier label enters inference. The environment retains hand-defined long multiplication sequencing; the learner chooses among 100 numerical claims. This domain scaffold is a limitation, not discovery of arbitrary procedures. The new head removes seven dummy policy losses and the artificial skip/sign score coupling. The output capacity is identical; only active supervision heads differ by task.

A new seed-5151 dataset contains 1000 training, 100 validation, 200 IID and 300 structural OOD examples. Structures remain train 1x1/2x1/2x2 and OOD 3x2/3x3/4x2. No test-based hyperparameter selection. Budgets 10/25/50/100/250/500/1000; seeds 0/1/2. Fixed 1200 Adam updates, batch64, lr0.003; final checkpoint. Data beyond1000 is deferred to avoid expanding scope to chase targets. Validation is diagnostic, never selection. All original test sets remain historical.

B receives teacher transition labels. C receives only its own accepted proposals and bounded scalar feedback (64 attempts/problem), followed by replay. A/structured receive final-answer labels. A matched-information control trains the same procedural backbone/head on B's exact local state/action rows without episodic or dream components; it is a conventional local supervised policy, not an answer-only learner receiving hidden labels. It establishes whether the wrapper adds anything beyond the representation/supervision. Costs report rows, consumed minibatch rows, optimizer steps, transformer input tokens/model calls, scalar feedback attempts, time and shared extraction cost separately.

Primary dream is disabled to avoid confidence lookahead cost; a bounded depth2/branch3 intervention measures it at n=50. No-state retrains B at n=50; no-memory matches C's accepted samples and optimizer update count using in-memory arrays instead of SQLite replay. No interaction performs proposed-state recurrence, with hypothetical evidence only. Goal evaluator removal is a diagnostic scoring removal, not a changed arithmetic test; no goal oracle exists in the policy. No-process-supervision uses the C regime and is explicitly a supervision/query contrast. All primary seeds are retained. Single-budget ablations cannot estimate delta N*.

Shared evaluator reports exact answer, verified trajectory, terminal goal and groundedness separately. Baseline trajectories are absent and validity is null. Groundedness comparisons inherit this output-protocol difference. N* uses observed mean-accuracy thresholds 0.8/0.9/0.95, no interpolation; missing crossings and DER are not_reached. Mean±sample SD, per-run Wilson95 intervals and paired exploratory McNemar at

n=1000 seed 0 with Holm correction; no seed pooling or claim from significance alone. Errors group by condition/budget/split/digit structure/carry count. Only actual frozen runs enter results/final.

The checkpoint revision is 12fd25f77366fa6b3b4b768ec3050bf629380bac. Actual frozen backbone134,515,008 parameters; trainable head337,618; total134,852,626. CPU execution because the installed runtime cannot access MPS. The categorical numeric feature adapter performs indexing/quantization only; it never multiplies digits or computes labels. The text encoder sees indexed numerical features, not natural instruction prompts. This limits interpretation of language/math pretraining benefits. All conditions share this adapter and head capacity.

No-goal-evaluator intervention removes the constant model-visible goal flag; external scoring remains unchanged. It tests that feature's contribution, not removal of benchmark correctness. The matched-information policy bypasses working/concept memory while preserving the exact learned predictor, local inputs and real execution evidence. No-memory arrays are in the same chronological order as episodic replay and use exactly1200 updates: it tests storage/retrieval, not the benefit of reuse itself. Cache reuse is shared and target-free; warm/cached latency is not presented as cold deployment latency or compute superiority. Tokens actually processed by the frozen transformer and feature requests are reported separately from gradient-update rows.

Final Limitations

PDF edition of preserved repository document | docs/final_limitations.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Completion limitations

Flagship C locked episodes: 100/100. Lockbox: 400/400. Learning: 1920/1920. Dedicated latency: 16/16. Counts describe actual saved evidence; completion requires every mandatory scope. The historical emergency package preserves its cutoff and failures. The authorized continuation removes that operational cutoff without changing the frozen scientific protocol.

Mixed CPU float32/MPS reduced precision; missing zero control; unmatched supervision and inference interaction budgets; synthetic bounded tasks; unknown foundation pretraining exposure; no learned world model; no recovery or component ablation results; tiny descriptive latency sample; public protocol mismatch; SQL model study deferred. Unmeasured results stay inconclusive. Hardware-only speedup is not isolated from batching and precision.

Final Metrics

PDF edition of preserved repository document | docs/final_metrics.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Actual completion metrics

Machine-readable full metrics: results/coding/emergency_analysis_v1/summary.json.

Mindscape bounded MPS completion evidence

Learning curve: 1920/1,920 episodes, 0 not run. Lockbox: 400/400 condition-task episodes (100 independent tasks). Preserved CPU work remains intact.

All metrics below derive from saved episodes. Missing cells are not zero scores. CPU float32 and MPS reduced-precision strata are kept separate. Supervision and interaction budgets differ across conditions; architecture-only causality is not established.

Condition	Split	Completed/planned	Success	95% Wilson interval
model_only	test	50/50	16.0%	[0.0833742067803340 4, 0.285142160630349 9]
model_only	ood_test	50/50	24.0%	[0.1429739139699173, 0.3741268375794292]
structured	test	50/50	100.0%	[0.9286524008666414, 1.0]
structured	ood_test	50/50	44.0%	[0.3116219921125717, 0.5769397197834315]
mindscape_b	test	50/50	90.0%	[0.7863976856252035, 0.9565242350681096]
mindscape_b	ood_test	50/50	48.0%	[0.3479713528657804 6, 0.614882551099553 9]
mindscape_c	test	50/50	98.0%	[0.8950455641036219, 0.9964607407283539]
mindscape_c	ood_test	50/50	36.0%	[0.2413874965184674 1, 0.498589831238873 07]

Observed complete-cell sample thresholds and all partial cells are in results/coding/emergency_analysis_v1/summary.json. DER is undefined for a causal architecture comparison.

No zero-budget control was run.

Task-paired confidence intervals are descriptive and unadjusted for multiple comparisons. Transition validity is conditional on recorded actions; malformed model proposals are reported separately.

Observed self-correction from saved offline first-edit assessments: [{"condition": "mindscape_b", "split": "test", "completed_episodes": 50, " first_edit_failures": 0, "later_terminal_successes": 0, "conditional_self_correction_rate": null, "wilson_ci95": null, "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_b", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 38, " later_terminal_successes": 12, "conditional_self_correction_rate": 0.3157894736842105, "wilson_ci95": [0.1908460367559618, 0.4745575989060954], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "test", "completed_episodes": 50, " first_edit_failures": 1, "later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.7934506856227626], " definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 32, " later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.1071791982550706], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}]. This is separate from unrun injected-fault recovery tests.

Operational answer metrics: {"definition": "Successful terminal program AND independent full-trajectory replay agreement; no semantic hallucination detector", "denominator": 400, "grounded_successful_answers": 228, "terminal_incorrect_answers": 172, "semantic_hallucination_rate": null, "semantic_hallucination_status": "Not measured by this program-repair protocol", "unverified_episodes": 0, "grounded_successful_answer_rate": 0.57, "unsupported_by_terminal_tests_rate": 0.43}. Failed terminal tests are unsupported program answers under this oracle; they are not a measured semantic hallucination rate.

Original frozen configuration SHA-256: 11d2a6b5bef6517a5cae10dca46f1e94bfe2db3504afd3942434fcb78f496bb0. Seeds: 11/23/37 for learning; seed 11 and budget 100 for lockbox. Exact checkpoint/source hashes are in the frozen protocol; host/runtime and training compute are in the machine-readable summary.

Actual acceleration measurements


```
{
  "cpu_seconds_per_episode": 111.3537078230629,
  "cpu_reference_scope": "Preserved fresh CPU float32 episodes; not rerun",
  "mps_throughput_seconds_per_episode": 27.54043339325017,
  "mps_mean_episode_wall_seconds": 372.8890333698155,
  "throughput_speedup": 4.043280882077708,
  "mps_episodes_per_hour": 130.7168971742586,
  "selected_batch_size": 16,
  "precision": "float16",
  "load_seconds": 22.024639666007715,
  "peak_process_rss_bytes": 3421274112,
  "mps_allocated_bytes": 3333364480,
  "mps_driver_bytes": 12526354432,
  "profiling_seconds": 600.09845495224,
  "interpretation": "Historical CPU vs fresh batched MPS throughput; per-case latency
reported separately. Arithmetic precision differs; learning-curve hardware strata remain
disclosed."
}
```


One phase-boundary memory release: {"before_driver_bytes": 10687807488, "after_driver_bytes": 4346019840, "live_allocated_bytes": 3112987904, "reason": "Single profiling/evaluation boundary; no per-operation cache clearing", "configuration_reused": true, "time_unix": 1791468859.762655}.

CPU reference episodes were previously measured with uncached inference; they were not rerun. MPS sec/episode is amortized throughput, distinct from per-episode latency while waiting for batches. Device tensors and model placement are checked; unsupported MPS operations fail with fallback disabled. Native allocation reported by the user is verified, but inference use is only established by native run outputs.

Replay and latency


```
{
  "replay": {
    "episodes": 400,
    "planned": 400,
    "final_outcome_agreement": 1.0,
    "structured_evidence_agreement": 1.0,
    "state_agreement": 1.0,
    "state_comparisons": 2584,
    "tool_result_agreement": 1.0,
    "tool_result_comparisons": 1092
  },
  "latency": [
    {
      "condition": "model_only",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 4
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3132599552,
      "peak_mps_driver_bytes": 4389109760,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1475.2094699714482,
      "percentiles": {
        "total": {
          "p50": 2.5287001245001193,
          "p95": 2.9951699749499765,
          "n": 4
        },
        "ttft": {
          "p50": 0.10764597949992094,
          "p95": 0.8070860983006238,
          "n": 4
        },
        "response": {
          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
        },
        "model_generation": {
          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.00013612449993161135,
          "p95": 0.0004516401001183111,
          "n": 4
        },
        "verification": {
          "p50": 0.3319560420000016,
          "p95": 0.3436878607001745,
          "n": 4
        },
```

```
        "test_execution": {
          "p50": 0.3318179789998794,
          "p95": 0.34356011874992876,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "structured",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 4
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3134778624,
      "peak_mps_driver_bytes": 4431052800,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1252.859663165493,
      "percentiles": {
        "total": {
          "p50": 2.891284770499624,
          "p95": 3.226074864299926,
          "n": 4
        },
        "ttft": {
          "p50": 0.3468195834998369,
          "p95": 0.3796293562498249,
          "n": 4
        },
        "response": {
          "p50": 2.5587374795004507,
          "p95": 2.895561100049872,
          "n": 4
        },
        "model_generation": {
          "p50": 2.5587374795004507,
          "p95": 2.895561100049872,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.0006867300003250421,
          "p95": 0.0008416026992108527,
          "n": 4
        },
        "verification": {
          "p50": 0.3316854585004876,
          "p95": 0.33450531509984105,
          "n": 4
        },
        "test_execution": {
          "p50": 0.33154877100014346,
          "p95": 0.3343919018002907,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "mindscape_b",
```

```
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 32
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3136957696,
      "peak_mps_driver_bytes": 4506550272,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 95.38661046707223,
      "percentiles": {
        "total": {
          "p50": 35.924512353999944,
          "p95": 44.30880428300006,
          "n": 4
        },
        "ttft": {
          "p50": 1.1655523545000506,
          "p95": 1.6721938956499798,
          "n": 32
        },
        "response": {
          "p50": 4.590948958500576,
          "p95": 5.841075649850472,
          "n": 32
        },
        "model_generation": {
          "p50": 35.228675708500305,
          "p95": 43.44583151340143,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.0012617284996849776,
          "p95": 0.0015499980499498632,
          "n": 4
        },
        "verification": {
          "p50": 0.3748883539997223,
          "p95": 0.47029650980048243,
          "n": 4
        },
        "test_execution": {
          "p50": 0.7139516665001793,
          "p95": 0.8574064588998226,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "mindscape_c",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 32
      },
      "actual_batch_sizes": [
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      ],
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      "peak_mps_live_bytes": 3139136768,
      "peak_mps_driver_bytes": 4514938880,
      "peak_cuda_allocated_bytes": null,
```

```
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 117.28219079755843,
      "percentiles": {
        "total": {
          "p50": 31.74315270850002,
          "p95": 32.581375693700465,
          "n": 4
        },
        "ttft": {
          "p50": 1.1719996460001312,
          "p95": 1.6940761644004851,
          "n": 32
        },
        "response": {
          "p50": 3.9009514585000034,
          "p95": 4.31798346860005,
          "n": 32
        },
        "model_generation": {
          "p50": 31.05320908349904,
          "p95": 31.83074219474984,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.0014917294993210817,
          "p95": 0.0035893313990072776,
          "n": 4
        },
        "verification": {
          "p50": 0.3700127080001039,
          "p95": 0.38461931249967163,
          "n": 4
        },
        "test_execution": {
          "p50": 0.6868583330001456,
          "p95": 0.7454004454003552,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    }
  ]
}
```


Preserved public benchmark

0.5B: 94/164 (57.32%); 1.5B: 98/164 (59.76%). Greedy one-sample full-module generation, 512-token cap, CPython 3.14.7 WASI. Foundation pretraining exposure is unknown. These local results do not exceed historical GPT-4 67.0% or Gemini Ultra 74.4%, and protocols differ. See docs/historical_references.md for pinned primary sources.

Dedicated latency: 16/16 fresh sequential cases; actual batch sizes and memory are reported above. Sequential response latency is distinct from frozen batch-16 evaluation throughput.

Complete suite: {'tests': 136, 'failures': 0, 'errors': 0, 'skipped': 0, 'pytest_passed_tests': 130, 'subtests_passed': 6, 'seconds': 22.683}. Local fairness/leakage assertions: 10158, PASS. Contamination-free status remains unknown. All 56 retained adapters validated. SQL environment implemented; SQL model study deferred. Recovery and memory/dream ablations not run; their claims remain inconclusive.

Final Package Verification

PDF edition of preserved repository document | docs/final_package_verification.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final coding package verification

Release acceptance requires exactly 400 locked rows, 1920 learning rows, 400 independent trajectory/terminal replays and 16 dedicated latency cases, passing complete tests, scoped lint, compile and wheel build. The publication gate checks exact task/key sets, byte-preserved CPU evidence, frozen source hashes and agreement between raw scores and reports.

The immutable snapshot is results/final/coding_research_v2. Its manifest binds each file to SHA-256 and records the source commit. Clean-destination restoration, inventory/hash verification and the restored full test suite are required before the completion tag. The post-snapshot verification receipt is distributed alongside the snapshot at results/final/coding_research_v2_verification.json and committed in experiments/coding_completion_v2/research_package_verified.json; it necessarily postdates the immutable snapshot. Foundation weights are exact pinned external downloads.

The historical partial snapshot coding_emergency_partial_v1 retains its original 6178 files and verification receipt. It is historical evidence, not the current completion scope.

Final Report

PDF edition of preserved repository document | docs/final_report.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final build completion report

IMPLEMENTED

Stronger frozen local backbone adaptation, categorical task representation, four regimes, procedural100-candidate execution, posthoc verification, working/episodic/concept memory, bounded dreams, generic environment protocol, immutable result roots, statistics/failures/ten figures and an evidence UI.

AUDITED

Historical encodings, targets, losses, optimization, splits, initialization/decoding, information and cost; executed sanity checks; final438-check artifact audit.

FIXES APPLIED

Removed dummy policy-head losses and artificial skip/sign coupling in the final adapter; categorical numerical representation and a larger head; matched replay ordering/update budget; strict checkpoint identity/reload tests; separated real/hypothetical posthoc annotations and resource units. Historical code/results retained.

MODELS USED

Pinned Apache2.0 SmolLM2-135M-Instruct frozen transformer plus learned task heads; historical NumPy MLP; deterministic ceiling, operand memorization and random references.

MODEL SIZES

134,515,008 frozen backbone +337,618 trainable head =134,852,626 parameters per final primary condition. Historical MLP8722; deterministic/memorization/random references0 neural parameters.

TRAINING REGIMES

Answer-only, structured answer-only, trajectory-supervised B, bounded binary-feedback/accepted-experience replay C. Same frozen backbone/head capacity; fixed1200 Adam updates/batch64. Labels and query amounts differ and are disclosed.

DATASETS

Fresh final claims dataset seed 5151:1000 train/100 validation/200 IID/300 OOD. Train1x1/2x1/2x2; OOD3x2/3x3/4x2. Canonical identity leakage checks; old development/research datasets preserved.

DATA BUDGETS

10,25,50,100,250,500,1000;2500/5000 deferred.

SEEDS

0,1,2 for all primary condition/budget pairs and component controls.

IID RESULTS

At1000 mean±SD:A5.000%±0.866; structured5.333%±1.041; B41.500%±2.179; C19.167%±3.014.

OOD RESULTS

At1000:B6.111%±1.347; A/structured/C0%±0. All OOD figures remain separate from IID.

DATA-EFFICIENCY RESULTS

Every N*(0.8/0.9/0.95) and DER is not reached; no2×/5× measured efficiency claim.

GROUNDEDNESS

IID at 1000:B41.500%±2.179; C19.167%±3.014; answer baselines0% by evidence-required protocol. OOD B6.111%±1.347; others0%.

TRAJECTORY VALIDITY

IID at 1000:B41.500%±2.179; C19.167%±3.014; OOD B6.111%±1.347/C0%; baselinesN/A.

GOAL SUCCESS

Same numerical rates as groundedness in the measured final primary runs; below95% target.

ABLATIONS

Three seeds at 50. No state:–3pp IID; C supervision contrast:–2.5pp IID; no real environment:0pp accuracy and–3pp grounded/goal by definition. Matched local policy, replay-store bypass, goal-flag removal and dream/no-dream:0pp accuracy. All OOD deltas0. ΔDER unavailable from a single budget.

DREAM RESULTS

Goal-directed tagged integer rollout demonstrated; multiplication depth2/branch3 adds calls but no measured accuracy/ground/goal change at 50. No learned world model or utility claim.

SECOND-VERTICAL STATUS

Deferred learned second environment. Generic interface and simulator reuse demonstrated; full learned architecture transfer not established.

HISTORICAL GPT/GEMINI REFERENCE STATUS

Published GPT-4/Gemini GSM8K sources retained with prompting/training caveats. Direct multiplication comparison unsuitable; no GPT superiority claim.

COMPUTE COST

Main:90 trained heads/210 evaluations; training171.947s/evaluation886.511s cached CPU;108,000 updates/6,912,000 sampled rows/379,153 feedback queries/303,924 inference-head calls;1,961,882 shared encoder tokens. Old-model/reference runs reported separately. No compute-efficiency superiority.

FAILURE ANALYSIS

22,367 arithmetic errors;28,762 wrong-action outcomes;133 unsupported outcomes;1238 grounded correct outcomes across210 repeated-set evaluations. Other listed failure classes0. Grouped digit/carry/condition/budget/split tables and example-level failures retained.

CLAIMS SUPPORTED

Narrow descriptive regime findings: B/C higher IID than answer-only under the tested unequal-supervision protocols; B nonzero verified OOD. Supervised pipeline can memorize; artifact/evidence separation verified. No broad architectural superiority claim is marked supported.

CLAIMS NOT SUPPORTED

Threshold-based data efficiency, replay-storage contribution, dream usefulness, second learned-domain transfer. Accuracy/OOD/grounding/trajectory/state causal claims are inconclusive under the available controls.

KNOWN LIMITATIONS

Hand-designed decomposition; unequal labels/queries; frozen indexed-feature backbone; one dataset/domain; small seed count; no independently reserved carry-only split; evidence metric privileges trajectory emitters; warm/shared CPU cache; second-domain and full fine-tuning deferred.

TEST COUNT

92 tests passed, preserving all 86 previous tests and adding six final regressions. Editable install, real checkpoint CLI, generator, training/reload and analysis/audits executed. Browser checked baseline toggle, real failed step and plots.

GIT STATUS

Final release on main with clean tracked working tree; historical commits preserved. Results/caches ignored; committed summary/figures/demo retained.

LATEST COMMIT

Resolve the stable final release with git rev-parse mindscape-final-study-v1; protocol records training-source commit5cbda01. Release hash is provided in the completion message.

FINAL REPOSITORY TREE

```
mindscape/
■■■ README.md / pyproject.toml / Makefile
■■■ src/mindscape/
■   ■■■ core/ data/ environments/ verification/
■   ■■■ models/ training/ evaluation/
■   ■■■ memory/ reasoning/ concepts/
■■■ configs/experiments/ [historical + final protocols]
■■■ datasets/generated/ [development_v1, research_v1, final_v1]
■■■ scripts/ [generation, training, evaluation, analysis, audit, freeze, demo]
■■■ tests/ [five suites,92 tests]
■■■ docs/ [architecture, protocol, results, conclusions, limits, reproduction]
■■■ experiments/ [historical snapshots, final_audit, final_smollm2_v1]
■■■ results/final/ [smollm2_v1, historical_mlp_v1, references_v1]
■■■ demo/ [index.html, screenshot.png]
■■■ work/ [ignored local runtime, pretrained weights, frozen-feature cache, logs]
```

Detailed results (final_results.md) · Reproduce (final_reproducibility.md) · Limitations (limitations.md)

Final Reproducibility

PDF edition of preserved repository document | docs/final_reproducibility.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Reproduce the final study

Python3.11+ and a local machine with sufficient memory for a135M frozen model. Tested Python3.12.14, NumPy2.3.5, PyTorch2.14.1, Transformers4.57.6 on Apple M5/16GB, CPU. No paid service or API key. Initial package/model downloads need internet; subsequent generation, adaptation and evaluation work offline. Exact installed optional package versions are in experiments/final_audit/local_requirements.txt. ReportLab renders scientific SVG charts.

Run from repository root:

```
python3 -m venv .venv
. .venv/bin/activate
python -m pip install -e '.[learning,plots,pretrained]'
PYTHONPATH=src python -m unittest discover -s tests -v
python scripts/download_final_model.py
```

The downloader prints the pinned snapshot path. Use that path below. Local weights are Apache2.0, from https://huggingface.co/HuggingFaceTB/SmolLM2-135M-Instruct . Backbone revision 12fd25f77366fa6b3b4b768ec3050bf629380bac. The head is trained locally. Cache folders are ignored by Git; do not mistake bundled results for bundled pretrained weights.

```
PYTHONPATH=src python scripts/generate_data.py \
  --config configs/experiments/final_dataset.toml \
  --output datasets/generated/final_reproduction
PYTHONPATH=src python scripts/run_final_study.py \
  --dataset datasets/generated/final_reproduction \
  --model work/hf/hub/models--HuggingFaceTB--
SmolLM2-135M-Instruct/snapshots/12fd25f77366fa6b3b4b768ec3050bf629380bac \
  --output results/final/reproduction_v1
PYTHONPATH=src python scripts/analyze_final_study.py results/final/reproduction_v1
PYTHONPATH=src python scripts/audit_final_artifacts.py results/final/reproduction_v1 \
  --dataset datasets/generated/final_reproduction
PYTHONPATH=src python scripts/final_references.py \
  --dataset datasets/generated/final_reproduction --output
results/final/references_reproduction
PYTHONPATH=src python scripts/build_final_demo.py results/final/reproduction_v1 \
  --dataset datasets/generated/final_reproduction --output demo/reproduction.html
```

All output roots must be new. The frozen study fails rather than overwriting an existing result. The full command adapts84 primary models and6 ablation models, reloads every checkpoint and evaluates each condition on IID/OOD. It may take substantial CPU time; the transformer feature cache amortizes repeated target-free state encodings. A small smoke study can use a new TOML with budgets [10], seeds [0], steps1200 and attempts_per_problem64; currently the full runner's ablation stage expects n50, so use budgets [10,50] when running that stage. Main final analyses require the declared complete three-seed/seven-budget matrix.

Evaluate one final checkpoint on a fresh output root:

```
PYTHONPATH=src python scripts/evaluate_final.py \
  --model work/hf/hub/models--HuggingFaceTB--
SmolLM2-135M-Instruct/snapshots/12fd25f77366fa6b3b4b768ec3050bf629380bac \
```

```
  --checkpoint results/final/reproduction_v1/full_trajectory_n1000_s0/checkpoint \
  --dataset datasets/generated/final_reproduction \
  --split ood_test --output results/rechecks/ood_v1
```

Inspect predictions.jsonl in each UUID evaluation folder. Fields include predicted/target answer, complete predicted trajectory, correctness, goal, groundedness and diagnostics. Correct-action/transition judgments are posthoc evaluation annotations; model inputs exclude them. Configs contain exact training IDs, regime and model metadata. Saved heads use NPZ without pickle. Ten summary SVG figures and per-run outcome plots are generated from measured runs. statistics.json, difficulty_failures.json, failure_taxonomy.json, data_efficiency.json and claims.json are machine-readable.

Reproduce the historical diagnostics with scripts/audit_final.py on a new copy or rename its new output directory: it intentionally refuses overwrites. Preserve original results rather than deleting them. Dream demonstration: PYTHONPATH=src python scripts/run_dream_demo.py. View the static evidence UI by opening demo/index.html or serving demo/ with python -m http.server 8000 --directory demo.

Free external compute is optional, not used in this study. No external scheduler/service is required. Reproductions have new timestamps and measured runtime; exact predictions should match with the pinned checkpoint, seed, CPU implementation and package versions, while other hardware/kernel versions may differ numerically.

Additional checks/references:

```
PYTHONPATH=src python scripts/final_sanity.py results/final/reproduction_v1 \
  --model work/hf/hub/models--HuggingFaceTB--
SmolLM2-135M-Instruct/snapshots/12fd25f77366fa6b3b4b768ec3050bf629380bac \
  --dataset datasets/generated/final_reproduction
PYTHONPATH=src python scripts/analyze_holdout_coverage.py results/final/reproduction_v1 \
  --dataset datasets/generated/final_reproduction
PYTHONPATH=src python scripts/final_historical_mlp.py \
  --dataset datasets/generated/final_reproduction --output
results/final/historical_mlp_reproduction
```

The last reference repeats the historical MLP at1000 problems, three seeds, on the fresh final dataset. It does not equalize its architecture, losses or update budget with the final frozen-backbone adaptation, so their difference does not isolate pretraining benefits.

Finalize measured resource reporting and digest the complete validated study:

```
PYTHONPATH=src python scripts/summarize_final_costs.py results/final/reproduction_v1
PYTHONPATH=src python scripts/freeze_final_artifacts.py results/final/reproduction_v1
```

The SHA256 freeze manifest covers raw predictions, checkpoints, metadata, statistics and plots. Generate all analyses and audits before freezing. Do not modify that folder afterward; use a new output root for a reproduction or correction.

Before final freezing, preserve separate hypothetical posthoc judgments without rewriting raw predictions:

```
PYTHONPATH=src python scripts/annotate_final_hypotheticals.py
results/final/reproduction_v1
```

Record the pinned downloaded model-file digests and freeze the separately validated references:

```
PYTHONPATH=src python scripts/model_manifest.py \
  --model work/hf/hub/models--HuggingFaceTB--
SmolLM2-135M-Instruct/snapshots/12fd25f77366fa6b3b4b768ec3050bf629380bac \
  --output experiments/reproduced_model_manifest.json
```

```
PYTHONPATH=src python scripts/freeze_final_references.py
results/final/historical_mlp_reproduction \
  --dataset datasets/generated/final_reproduction
PYTHONPATH=src python scripts/freeze_final_references.py
results/final/references_reproduction \
  --dataset datasets/generated/final_reproduction
```

Before freezing the main study, optionally render the retained readable ablation variant with python scripts/final_ablation_plot.py results/final/reproduction_v1. The original figure remains preserved. scripts/write_final_report.py generates the thesis-ready snapshot for this repository's declared study paths; for a new reproduction, use its measured JSON inputs with new snapshot paths rather than overwriting the committed report. The exact used global/local package versions and CPU hardware are also recorded in experiments/final_audit/runtime.json.

Final Research Report

PDF edition of preserved repository document | docs/final_research_report.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final frozen coding-study report

Verified coverage: 1920 learning / 400 lockbox / 400 independent audits / 16 fresh latency cases. Original1634-row checkpoint and original1122CPU-row prefix remain byte-identical. Recovery archive is preserved. Neural PID1620 exited; supervisor completed every checked subprocess.

Observed outcomes

Condition	Split	Successes /50	95% Wilson interval
model_only	test	8/50	0.083–0.285
model_only	ood_test	12/50	0.143–0.374
structured	test	50/50	0.929–1.000
structured	ood_test	22/50	0.312–0.577
mindscape_b	test	45/50	0.786–0.957
mindscape_b	ood_test	24/50	0.348–0.615
mindscape_c	test	49/50	0.895–0.996
mindscape_c	ood_test	18/50	0.241–0.499

Task-paired statistics and complete learning cells are in experiments/coding_completion_v2/final_summary.json. Intervals are descriptive and unadjusted for multiple comparisons. Successful replay verifies evidence, not successful repair. Five zero-tool trajectories have undefined tool-result agreement.

All dedicated latency observations

Sequential, cache-disabled MPSfloat16; no model reload per case. Stage intervals can overlap. Four task/condition pairs per condition; no repeated key counted twice.

Condition	Task	Split	Success	Wall seconds
model_only	fd23fa32e4f9	test	False	3.030
model_only	e5545f5d719e	ood_test	False	2.800

Condition	Task	Split	Success	Wall seconds
model_only	34d9c496b646	test	False	2.257
model_only	f7f984138799	ood_test	False	1.674
structured	fd23fa32e4f9	test	True	2.470
structured	e5545f5d719e	ood_test	False	3.142
structured	34d9c496b646	test	True	2.641
structured	f7f984138799	ood_test	False	3.241
mindscape_b	fd23fa32e4f9	test	True	33.891
mindscape_b	e5545f5d719e	ood_test	False	45.430
mindscape_b	34d9c496b646	test	True	37.958
mindscape_b	f7f984138799	ood_test	False	33.686
mindscape_c	fd23fa32e4f9	test	True	26.679
mindscape_c	e5545f5d719e	ood_test	False	32.615
mindscape_c	34d9c496b646	test	True	31.098
mindscape_c	f7f984138799	ood_test	True	32.388

Validation and reproducibility

Fresh original and clean-restored suites each passed 130tests plus6subtests, zero failures/errors/skips. Scoped lint, compilation, wheel build, fairness/training checks and Nodeplayback verification passed. Snapshot results/final/coding_research_v2 and restored file hashes independently verified:6242 files each. Full commands/outcomes, all16 stage measurements, scientific pins and SHA256checksums are in experiments/coding_completion_v2/release_manifest.json.

Limitations

- Mixed hardware/precision and unmatched supervision/interaction compute prevent architecture-only causal attribution

- Synthetic bounded repair tasks; unknown foundation pretraining exposure

- 16latencycases are descriptive, not broad performance estimates

- No measured zero-budget/control component memory/dream/recovery intervention

- Foundation weights are pinned external downloads, excluded from package

- No configured type-check acceptance task; no visual browser QA claimed

- No GPT/Gemini superiority; negative and unsuccessful cases retained

Completion refers to frozen experimental coverage. Broad capability and GPT/Gemini superiority are unsupported; data-efficiency and unmeasured intervention claims remain inconclusive. Package publication/tag verification is recorded separately after export.

Final Results

PDF edition of preserved repository document | docs/final_results.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final results

Final main study:90 adapted heads (84 primary plus 6 retrained ablations),210 shared-runner evaluations, three seeds 0/1/2,200 IID and300 OOD examples per run. An additional12 historical MLP models/24 evaluations and6 non-neural reference evaluations use the same fresh final dataset. Total final research evaluations240;102 trained research models. Diagnostics are separate. Raw results remain in results/final/; the committed summary snapshot is experiments/final_smollm2_v1/.

Primary results at 1000 training problems

Condition	IID exact accuracy	Structural OOD exact accuracy	IID groundedness	IID goal success	IID trajectory validity
Answer-only A	5.000% ± 0.866	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	N/A
Structured baseline	5.333% ± 1.041	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	N/A
Mindscape B	41.500% ± 2.179	6.111% ± 1.347	41.500% ± 2.179	41.500% ± 2.179	41.500% ± 2.179
Mindscape C	19.167% ± 3.014	0.000% ± 0.000	19.167% ± 3.014	19.167% ± 3.014	19.167% ± 3.014

All rates are mean±sample SD across three training seeds. A/structured do not emit trajectories; validity isN/A and their evidence-required grounding/goal scores are0 by protocol. B OOD groundedness/goal/validity are6.111%±1.347; C OOD rates are0. No final success target was met.

Learning curves

IID

Training problems	Answer-only A	Structured	B	C
10	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000
25	0.167% ± 0.289	0.000% ± 0.000	2.000% ± 0.000	0.500% ± 0.866
50	0.333% ± 0.289	0.167% ± 0.289	3.000% ± 1.323	0.500% ± 0.500
100	0.667% ± 0.289	0.667% ± 0.289	13.333% ± 1.041	2.667% ± 2.517
250	2.333% ± 0.577	0.833% ± 0.577	30.333% ± 1.041	9.667% ± 3.215

Training problems	Answer-only A	Structured	B	C
500	1.333% ± 0.764	2.333% ± 0.764	38.000% ± 0.866	15.000% ± 3.606
1000	5.000% ± 0.866	5.333% ± 1.041	41.500% ± 2.179	19.167% ± 3.014

Structural OOD

Training problems	Answer-only A	Structured	B	C
10	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000
25	0.000% ± 0.000	0.000% ± 0.000	0.111% ± 0.192	0.000% ± 0.000
50	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000
100	0.000% ± 0.000	0.000% ± 0.000	0.889% ± 0.192	0.000% ± 0.000
250	0.000% ± 0.000	0.000% ± 0.000	1.556% ± 0.694	0.111% ± 0.192
500	0.000% ± 0.000	0.000% ± 0.000	4.667% ± 0.333	0.111% ± 0.192
1000	0.000% ± 0.000	0.000% ± 0.000	6.111% ± 1.347	0.000% ± 0.000

Data efficiency and costs

All observed N*(0.8/0.9/0.95) thresholds were not reached on either split for every condition; all DERs are censored/not reached. There is no measured2× or5× threshold-crossing advantage. Problem budget is distinct from label, feedback-query and compute budget.

Main study measured training time 171.947s and evaluation time 886.511s. It performed108,000 optimizer updates,6,912,000 sampled supervision rows,379,153 scalar feedback queries and303,924 prediction-head calls. The shared frozen transformer processed1,961,882 tokens in23,463 batches; encoder feature requests726,352. These are cached CPU measurements, exclude dataset/artifact bookkeeping outside timed sections, and do not establish cold-deployment compute efficiency. Full per-model costs are in costs.json.

Ablations at 50 problems

Intervention	IID Δaccuracy (pp)	OOD Δaccuracy (pp)	IID Δgroundedness (pp)	IID Δgoal (pp)
dream	0.000	0.000	0.000	0.000
matched_information_control	0.000	0.000	0.000	0.000
no_dream	0.000	0.000	0.000	0.000

Intervention	IID Δaccuracy (pp)	OOD Δaccuracy (pp)	IID Δgroundedness (pp)	IID Δgoal (pp)
no_environment_interaction	0.000	0.000	-3.000	-3.000
no_episodic_memory	0.000	0.000	0.000	0.000
no_explicit_state	-3.000	0.000	-3.000	-3.000
no_goal_evaluator	0.000	0.000	0.000	0.000
no_trajectory_supervision_regime_C	-2.500	0.000	-2.500	-2.500

Every intervention has three seeds. Δdata efficiency is not estimable from one budget. No goal evaluator means removal of a constant goal input flag while retaining common external scoring. C versus B is a supervision/feedback contrast, not an isolated removal. No real interaction preserves numerical predictions but loses real evidence by definition. The matched local supervised policy and matched-order/update no-SQLite-retrieval control have identical predictions to their references. Dream changed no measured accuracy/groundedness/goal; its extra calls do not support usefulness. The no-state mean reduction is not significant in every seed after the component-test Holm family, so state-contribution claims remain inconclusive at this budget.

Statistics and errors

Per-run Wilson95 intervals report conditional example uncertainty; seed SD reports training variation. Exploratory paired tests use seed 0 at 1000, six Holm comparisons; component tests are a separate 18-comparison family across seeds/splits. No seed pooling. B versus A IID Holm p=2.2354e-17 and OOD p=1.90735e-6; C versus A IID p=1.80444e-7. These differences are observed regime associations, not controls for the unequal information/scaffold. Practical OOD performance remains6.11%, far below90%.

Failure/outcome category	Count across210 repeated-set evaluations
arithmetic_error	22367
wrong_action	28762
invalid_transition	0
malformed_state	0
trajectory_failure	0
goal_failure	0
unsupported_answer	133
timeout	0
other	0
correct	1238

These counts describe repeated benchmark runs, not unique independent errors. difficulty_failures.json groups condition/budget/seed/IID-OOD/digit structure/carry count, and per-run failures.json retains example

identities and original taxonomy. Main failures are wrong local claims and answer arithmetic errors; syntax/timeouts were not the limiting issue. At1000,108/200 IID examples and280/300 OOD examples contain reference local contexts absent from the training subset. Carry signatures are unseen for59/200 IID and300/300 OOD. The latter includes the structural length change; these are posthoc descriptive strata, not independently pre-reserved carry-only holdouts.

Sanity and interpretive references

The final supervised representation memorized n=10, all 45 canonical1x1 problems and a mixed100 training set at 100% in diagnostics. Mixed100 validation reached3% answer-only and18% B; local claim accuracy62.47%, carry label accuracy82.60%. C n=10 same-set accuracy was0% with incomplete accepted experience; it did not receive complete trajectories. This separates fitting ability from generalization and exploration coverage.

Historical MLP on fresh final dataset at 1000	IID mean±SD	OOD mean±SD
Answer-only A	0.333% ± 0.289	0.000% ± 0.000
Structured baseline	0.333% ± 0.289	0.000% ± 0.000
Mindscape B	8.000% ± 0.866	0.667% ± 0.000
Mindscape C	8.333% ± 2.021	0.333% ± 0.333

The original phase-4 dataset remains separate: its OOD accuracy was0% across all primary runs. Different sampled datasets can change rare correct counts. Larger final heads/categorical features/active losses/update budgets/pretrained features changed together, so gains over the MLP do not isolate pretraining. Deterministic execution scored 100% IID/OOD; operand memorization and the seeded uniform-integer random reference scored 0%. The deterministic ceiling is an algorithm, not a learning/data-efficiency baseline.

Claims

Claim	Status	Evidence	Limits
Mindscape improves accuracy	INCONCLUSIVE	See primary curves and matched-information control.	State/scaffold and supervision differ from answer baselines; wrapper causal benefit must exceed matched local policy.
Mindscape improves data efficiency	NOT SUPPORTED	Observed N* and DER only.	Unequal label/query counts; compute reported separately.
Mindscape improves structural OOD generalization	INCONCLUSIVE	Unseen structures evaluated separately.	One domain and hand-defined cursor.
Mindscape improves groundedness	INCONCLUSIVE	Verified executed evidence measured.	Answer-only protocol emits no trajectory; metric favors evidence emitters by definition.

Claim	Status	Evidence	Limits
Mindscape improves trajectory validity	INCONCLUSIVE	Complete verifier checks.	Baselines without trajectories have null validity; matched policy is required.
Explicit state contributes	INCONCLUSIVE	Retrained no-state intervention at n=50.	One budget and changes representation. Support, if present, is limited to IID at n=50; OOD is separately reported.
Episodic memory contributes	NOT SUPPORTED	Matched accepted rows/order/updates no-storage control.	Tests SQLite retrieval versus in-memory replay, not all reuse.
Dream simulation contributes	NOT SUPPORTED	Depth2/branch3 intervention n=50.	Confidence scoring, not learned world model; extra inference calls.
Architecture transfers to another environment	NOT SUPPORTED	Goal-directed generic integer dream demo retained.	No second learned benchmark; extension deferred.

Supported descriptive findings: under these tested unequal-supervision protocols, B/C have higher IID accuracy than answer-only, and B executes some verified OOD trajectories. These observations do not establish an independent Mindscape wrapper advantage; matched-information policy predictions are identical. Data-efficiency, replay-storage, dream usefulness and second-domain-transfer claims are not supported. Broader causal claims remain inconclusive.

Artifact audit:438 actual checks passed, including nested training IDs, shared parameter count, no singleton forcing, proposal/result separation, posthoc judgments, hypothetical tags and matched-control prediction equality.

Figures

- accuracy (../experiments/final_smollm2_v1/plots/accuracy.svg)

- ood_accuracy (../experiments/final_smollm2_v1/plots/ood_accuracy.svg)

- groundedness (../experiments/final_smollm2_v1/plots/groundedness.svg)

- goal_success (../experiments/final_smollm2_v1/plots/goal_success.svg)

- trajectory_validity (../experiments/final_smollm2_v1/plots/trajectory_validity.svg)

- data_efficiency (../experiments/final_smollm2_v1/plots/data_efficiency.svg)

- ablation_effects (../experiments/final_smollm2_v1/plots/ablation_effects.svg)

- errors (../experiments/final_smollm2_v1/plots/errors.svg)

- training_compute (../experiments/final_smollm2_v1/plots/training_compute.svg)

- inference_cost (../experiments/final_smollm2_v1/plots/inference_cost.svg)

<!-- emergency-mps-results -->

Mindscape bounded MPS completion evidence

Learning curve: 1920/1,920 episodes, 0 not run. Lockbox: 400/400 condition-task episodes (100 independent tasks). Preserved CPU work remains intact.

All metrics below derive from saved episodes. Missing cells are not zero scores. CPU float32 and MPS reduced-precision strata are kept separate. Supervision and interaction budgets differ across conditions; architecture-only causality is not established.

Condition	Split	Completed/planned	Success	95% Wilson interval
model_only	test	50/50	16.0%	[0.0833742067803340 4, 0.285142160630349 9]
model_only	ood_test	50/50	24.0%	[0.1429739139699173, 0.3741268375794292]
structured	test	50/50	100.0%	[0.9286524008666414, 1.0]
structured	ood_test	50/50	44.0%	[0.3116219921125717, 0.5769397197834315]
mindscape_b	test	50/50	90.0%	[0.7863976856252035, 0.9565242350681096]
mindscape_b	ood_test	50/50	48.0%	[0.3479713528657804 6, 0.614882551099553 9]
mindscape_c	test	50/50	98.0%	[0.8950455641036219, 0.9964607407283539]
mindscape_c	ood_test	50/50	36.0%	[0.2413874965184674 1, 0.498589831238873 07]

Observed complete-cell sample thresholds and all partial cells are in results/coding/emergency_analysis_v1/summary.json. DER is undefined for a causal architecture comparison. No zero-budget control was run.

Task-paired confidence intervals are descriptive and unadjusted for multiple comparisons. Transition validity is conditional on recorded actions; malformed model proposals are reported separately.

Observed self-correction from saved offline first-edit assessments: [{"condition": "mindscape_b", "split": "test", "completed_episodes": 50, " first_edit_failures": 0, "later_terminal_successes": 0, "conditional_self_correction_rate": null, "wilson_ci95": null, "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_b", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 38, " later_terminal_successes": 12, "conditional_self_correction_rate": 0.3157894736842105, "wilson_ci95": [0.1908460367559618,

0.4745575989060954], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "test", "completed_episodes": 50, " first_edit_failures": 1, "later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.7934506856227626], " definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 32, " later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.1071791982550706], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}]. This is separate from unrun injected-fault recovery tests.

Operational answer metrics: {"definition": "Successful terminal program AND independent full-trajectory replay agreement; no semantic hallucination detector", "denominator": 400, "grounded_successful_answers": 228, "terminal_incorrect_answers": 172, "semantic_hallucination_rate": null, "semantic_hallucination_status": "Not measured by this program-repair protocol", "unverified_episodes": 0, "grounded_successful_answer_rate": 0.57, "unsupported_by_terminal_tests_rate": 0.43}. Failed terminal tests are unsupported program answers under this oracle; they are not a measured semantic hallucination rate.

Original frozen configuration SHA-256: 11d2a6b5bef6517a5cae10dca46f1e94bfe2db3504afd3942434fcb78f496bb0. Seeds: 11/23/37 for learning; seed 11 and budget 100 for lockbox. Exact checkpoint/source hashes are in the frozen protocol; host/runtime and training compute are in the machine-readable summary.

Actual acceleration measurements


```
{
  "cpu_seconds_per_episode": 111.3537078230629,
  "cpu_reference_scope": "Preserved fresh CPU float32 episodes; not rerun",
  "mps_throughput_seconds_per_episode": 27.54043339325017,
  "mps_mean_episode_wall_seconds": 372.8890333698155,
  "throughput_speedup": 4.043280882077708,
  "mps_episodes_per_hour": 130.7168971742586,
  "selected_batch_size": 16,
  "precision": "float16",
  "load_seconds": 22.024639666007715,
  "peak_process_rss_bytes": 3421274112,
  "mps_allocated_bytes": 3333364480,
  "mps_driver_bytes": 12526354432,
  "profiling_seconds": 600.09845495224,
  "interpretation": "Historical CPU vs fresh batched MPS throughput; per-case latency
reported separately. Arithmetic precision differs; learning-curve hardware strata remain
disclosed."
}
```


One phase-boundary memory release: {"before_driver_bytes": 10687807488, "after_driver_bytes": 4346019840, "live_allocated_bytes": 3112987904, "reason": "Single profiling/evaluation boundary; no per-operation cache clearing", "configuration_reused": true, "time_unix": 1791468859.762655}.

CPU reference episodes were previously measured with uncached inference; they were not rerun. MPS sec/episode is amortized throughput, distinct from per-episode latency while waiting for batches. Device tensors and model placement are checked; unsupported MPS operations fail with fallback disabled. Native allocation reported by the user is verified, but inference use is only established by native run outputs.

Replay and latency

```
{
  "replay": {
    "episodes": 400,
    "planned": 400,
    "final_outcome_agreement": 1.0,
    "structured_evidence_agreement": 1.0,
    "state_agreement": 1.0,
    "state_comparisons": 2584,
    "tool_result_agreement": 1.0,
    "tool_result_comparisons": 1092
  },
  "latency": [
    {
      "condition": "model_only",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 4
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3132599552,
      "peak_mps_driver_bytes": 4389109760,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1475.2094699714482,
      "percentiles": {
        "total": {
          "p50": 2.5287001245001193,
          "p95": 2.9951699749499765,
          "n": 4
        },
        "ttft": {
          "p50": 0.10764597949992094,
          "p95": 0.8070860983006238,
          "n": 4
        },
        "response": {
          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
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          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.00013612449993161135,
          "p95": 0.0004516401001183111,
          "n": 4
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        "verification": {
          "p50": 0.3319560420000016,
          "p95": 0.3436878607001745,
          "n": 4
        },
        "test_execution": {
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          "p95": 0.34356011874992876,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
```

```
      "condition": "structured",
      "episodes": 4,
      "planned": 4,
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      "device_precision_counts": {
        "mps/float16": 4
      },
      "actual_batch_sizes": [
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      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3134778624,
      "peak_mps_driver_bytes": 4431052800,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1252.859663165493,
      "percentiles": {
        "total": {
          "p50": 2.891284770499624,
          "p95": 3.226074864299926,
          "n": 4
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        "ttft": {
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          "p95": 0.3796293562498249,
          "n": 4
        },
        "response": {
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          "p95": 2.895561100049872,
          "n": 4
        },
        "model_generation": {
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          "p95": 2.895561100049872,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.0006867300003250421,
          "p95": 0.0008416026992108527,
          "n": 4
        },
        "verification": {
          "p50": 0.3316854585004876,
          "p95": 0.33450531509984105,
          "n": 4
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        "test_execution": {
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          "p95": 0.3343919018002907,
          "n": 4
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      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "mindscape_b",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
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      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
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      "peak_mps_driver_bytes": 4506550272,
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```
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      "peak_cuda_reserved_bytes": null,
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          "p95": 44.30880428300006,
          "n": 4
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        "ttft": {
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          "p95": 1.6721938956499798,
          "n": 32
        },
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          "n": 32
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        "model_generation": {
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          "p95": 43.44583151340143,
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        "verification": {
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          "p95": 0.47029650980048243,
          "n": 4
        },
        "test_execution": {
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          "p95": 0.8574064588998226,
          "n": 4
        }
      },
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 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "mindscape_c",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 32
      },
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      ],
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      "peak_mps_live_bytes": 3139136768,
      "peak_mps_driver_bytes": 4514938880,
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      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 117.28219079755843,
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          "p95": 32.581375693700465,
          "n": 4
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          "n": 32
```

```
        },
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          "n": 32
        },
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          "p95": 31.83074219474984,
          "n": 4
        },
        "environment_execution": {
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          "p95": 0.0035893313990072776,
          "n": 4
        },
        "verification": {
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          "p95": 0.38461931249967163,
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        },
        "test_execution": {
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          "p95": 0.7454004454003552,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    }
  ]
}
```


Preserved public benchmark

0.5B: 94/164 (57.32%); 1.5B: 98/164 (59.76%). Greedy one-sample full-module generation, 512-token cap, CPython 3.14.7 WASI. Foundation pretraining exposure is unknown. These local results do not exceed historical GPT-4 67.0% or Gemini Ultra 74.4%, and protocols differ. See docs/historical_references.md for pinned primary sources.

Dedicated latency: 16/16 fresh sequential cases; actual batch sizes and memory are reported above. Sequential response latency is distinct from frozen batch-16 evaluation throughput.

Complete suite: {'tests': 136, 'failures': 0, 'errors': 0, 'skipped': 0, 'pytest_passed_tests': 130, 'subtests_passed': 6, 'seconds': 22.683}. Local fairness/leakage assertions: 10158, PASS. Contamination-free status remains unknown. All 56 retained adapters validated. SQL environment implemented; SQL model study deferred. Recovery and memory/dream ablations not run; their claims remain inconclusive.

Historical References

PDF edition of preserved repository document | docs/historical_references.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Historical external references: suitability audit

These published general-model references are not comparable with our procedural multiplication task. They are contextual records, not achieved Mindscape targets.

Exact checkpoint hashes are not publicly specified in the cited reports; keep them unknown rather than guessing an API snapshot.

Model	Report/version	Dataset/task	Published score	Protocol	Source	Publication date	Suitability
GPT-4	March 2023 technical report evaluation; exact checkpoint undisclosed	GSM8K, grade-school word problems	92.0%	5-shot chain-of-thought; report notes GSM8K training data in pretraining mix	[Table 2 and Appendix E](https://cdn.openai.com/papers/gpt-4.pdf)	Report first submitted 2023-03-15; linked PDF dated 2023-03-27	Not directly comparable
Gemini Ultra	Gemini 1.0, arXiv 2312.11805 v1; exact checkpoint undisclosed	GSM8K, grade-school word problems	94.4%	Chain-of-thought self-consistency, Table 2 labelled Maj1@32; exact shot/temperature not specified in this table	[Technical report, Table 2](https://arxiv.org/html/2312.11805v1)	2023-12-19 submission	Not directly comparable

GPT-4 date metadata: arXiv record (https://arxiv.org/abs/2303.08774).

Gemini date/version metadata: v1 record (https://arxiv.org/abs/2312.11805v1).

Checked against primary reports on 2026-10-08. No live proprietary API was used.

Why a direct claim would be invalid

GSM8K requires understanding word problems and multi-step mathematics. Mindscape's current inputs are two explicit integers with a hand-specified decimal procedure.

The tasks, inputs, training distributions, decoding and compute differ. Published pretraining sizes are not a controlled domain-example budget; a ratio against our 1000 multiplication examples would not be a valid DER. Do not place these scores

on our accuracy/data-efficiency plots or claim that multiplication accuracy matches or beats either system. The two historical protocols also differ from each other.

Methodology for a future genuine comparison

Choose a shared task/dataset/split first. Freeze interpretation, scoring, prompt examples, decoding/sample count, tool access and evaluation budget. Verify that Mindscape actually supports the task; do not carve an easier subset after seeing results. Cite exact published revision/protocol, record any unknown checkpoint or settings, audit contamination and separate reference-only from reproduced results. If exact protocol compatibility cannot be established, report both contextual values with explicit differences and make no superiority or training-data-efficiency claim. Use the internal four-condition controlled comparison as the primary evidence.

Limitations

PDF edition of preserved repository document | docs/limitations.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Limitations

The multiplication scaffold, cursor progression, terminal-state construction and digit/carry mechanics are hand defined. Models learn numerical claims within that scaffold. They do not discover long multiplication or demonstrate general-purpose reasoning. The core interface is reusable, but the current state schema contains multiplication fields; no second learned-domain transfer is established.

The final transformer is frozen. Its indexed numerical feature text differs from ordinary instruction prompting, and categorical features accompany its hidden representation. Improvements over the historical MLP cannot isolate pretraining, representation, capacity, loss masking or update budget. Its135M parameters make the final model larger, not necessarily a stronger arithmetic model; measured results must decide. All final conditions share its capacity.

Answer labels, trajectory labels and scalar experiential feedback carry different amounts of information. Each B problem can contribute several supervised rows; C uses up to64 feedback attempts/problem and may not complete an episode. Dataset budget is not label, token or compute budget. The matched local supervised policy controls information for a wrapper comparison. Data-efficiency ratios against answer-only learners cannot alone establish architecture efficiency.

Groundedness and goal-success metrics require emitted verified trajectories. Answer-only outputs therefore score0 for those metrics even when numerically correct; this is a protocol constraint, not evidence of intrinsic unreliability. Trajectory validity is null for absent trajectories. No-environment recurrence loses real evidence by definition and is not an isolated arithmetic capability effect. The no-goal intervention removes a constant model-visible flag; external evaluator correctness remains mandatory.

No-episodic-memory control uses the same accepted rows, order and update count in memory rather than retrieving SQLite replay. It tests this storage/retrieval mechanism, not all possible benefits of episodic reuse. Concept descriptions are stored but no measured concept-learning contribution is claimed. Dream uses proposed-state dynamics and confidence, not a trained world model. A small generic goal demo demonstrates simulation interfaces only.

Nonzero1x1 canonical pairs are exhausted in training; IID holdouts therefore comprise larger seen structures. Structural OOD holds out3x2/3x3/4x2; carry/structure novelty is analyzed posthoc, not every possible carry pattern pre-reserved. Eight output digits and six input digits bound this adapter. The signed/zero-capable environment is broader than the final nonzero positive training distribution.

Only three training seeds, one generated dataset and one domain are studied. Mean±sample SD measures seed variation; Wilson intervals describe conditional per-run example uncertainty and do not remove shared-test dependence. One-seed McNemar tests are exploratory, with Holm adjustment, not cross-seed proof. Absolute and practical performance matter.

CPU inference uses a target-free shared frozen-feature cache. Reported latencies are warm/cached research measurements; first misses depend on run order. Shared transformer work and actual tokens are reported separately. They do not establish cold latency or end-to-end compute superiority. Full fine-tuning, larger budgets and new domains were deferred rather than expanding scope to chase success.

Historical GPT-4/Gemini GSM8K references use different tasks, prompting and training data. Direct comparison is unsuitable. No paid service was used. Original failures and configs are retained.

Milestone 01

PDF edition of preserved repository document | docs/milestone_01.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

First milestone report

Implemented

Phase 0: Git main branch, installable Python 3.11+ package, MIT license, TOML configuration loader, documentation, Makefile, optional lint/type/test tools and a CI test workflow.

Phase 1: frozen Entity, State, Relation, Action, Event, Result, Observation, Goal, Transition and Trajectory records. Five distinct evidence categories.

Phase 2: deterministic decimal digit/carry multiplication, partial-row accumulation, signed inputs and zero, required environment API and trajectory recording.

Phase 3: full transition replay verification, independent final-product oracle and goal evaluation. Illegal actions are rejected without modifying the state.

Tested

Python 3.12.14 standard-library unittest runner: 21 tests passed. Exhaustive oracle checks cover all 1,681 pairs in $[-20,20]^2$; 200 seeded pairs cover larger signed operands. Also tested carries, shifted partial rows, input/action validation, immutability, reset, configuration, and tampered/reordered/duplicated/incomplete trajectories. Seeded oracle checks are tests, not generated benchmark datasets. pytest, Ruff and MyPy were not installed or run. CI configuration exists but its hosted execution has not been observed. No neural experiments were performed.

Verified demonstration: 19 × 3

Initial digits (least significant first): [9,1] and [3]. Position 0, carry 0, row value 0, accumulated value 0. Goal: produce the correct signed integer product.

Action	Event	Actual result	Position after	Carry after	Row after	Total after	Phase after
multiply	digit_written	27	1	2	7	0	multiply
multiply	digit_written	5	2	0	57	0	flush
flush	carry_flushed	57	2	0	57	0	accumulate
accumulate	row_accumulated	57	0	0	0	57	finish
finish	answer_materialized	57	0	0	0	57	done

First digit: 9×3+0=27 → write 7, carry 2. Second: 1×3+2=5 → row 57, carry 0. Flush carry, accumulate row and materialize answer. Final state: done, answer 57, accumulated 57, carry 0. Trajectory valid=true, final answer valid=true, goal reached=true. Every before/after state, action, event and actual result is saved in examples/19_times_3.json.

Not yet implemented

Dataset generation/splits and their reproducibility/OOD tests; benchmark metric calculations; model/backbone interfaces and baselines; learning regimes; memory; data-efficiency studies; operational ablations; historical reference collection; simulation; second domain; UI and multimodal components. Placeholders are scaffolding.

Known limitations

The policy is a prescribed algorithm with one valid next action; no learned procedure or cognitive superiority is demonstrated. Replay shares environment transition logic; the independent product oracle and broad arithmetic tests mitigate shared bugs. Input observation is authoritative and the verifier cannot authenticate an external user's original question. Only in-memory typed records are replayed; JSON import and schema migration are not implemented. Dataclass annotations are not universal runtime validation, though input boundaries and legal actions are validated. Large operands are subject to Python integer-string conversion and resource limits. Partial-row addition is an exact integer operation rather than an addition-domain

trajectory. Local runtime was 3.12; the configured 3.11 CI has not yet run.

Relevant repository tree

```
mindscape/
■■■ README.md, LICENSE, pyproject.toml, Makefile
■■■ configs/{base,multiplication}.toml
■■■ docs/{architecture,research_hypotheses,benchmark,experimental_protocol}.md
■■■ docs/decisions/001-foundation.md
■■■ examples/19_times_3.json
■■■ src/mindscape/core/{schema,config,entities,states,...}.py
■■■ src/mindscape/environments/base.py
■■■ src/mindscape/environments/multiplication/environment.py
■■■ src/mindscape/verification/verifier.py
■■■ scripts/run_episode.py
■■■ tests/test_foundation.py
■■■ .github/workflows/tests.yml
```

Git

Branch main. Logical commits:

- d3d2261 repository and research specification
- a991db1 typed schemas and configuration
- 963dec9 environment and verifier
- ac51554 tests and recorded demonstration

This report is committed in a final documentation commit. Final status is checked separately after committing.

Milestone 02

PDF edition of preserved repository document | docs/milestone_02.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Phase 2 milestone: dataset and benchmark infrastructure

Implemented

General benchmark schemas and backend registry; deterministic configurable generation; identity-aware splits, OOD isolation, checksum/target/leakage validation; three separate data forms; nested budgets; measurable difficulty filters; independent metrics/N*/DER; abstract model interface, failure records, runner, immutable artifacts and four CLI operations.

Tested and verified

54 tests passed on Python 3.12.14, including all 21 original tests. Tests cover generation, seeds/byte reproducibility, JSON schema roundtrips, targets, digit ranges, signed inputs, difficulty bounds, split identities, swapped OOD/operands, trace contamination, target-free views, subsets, metric denominators, missing/malformed/foreign/incomplete trajectories, N*/DER, exception records, pre-call validation, no overwrite and repeat-run predictions.

A non-arithmetic test-only backend verifies generic pipeline reuse.

Development dataset: train 1000; validation 250; IID test 500; OOD test 500.

All targets and leakage checks passed. Nested budgets: 10,25,50,100,250,500,1000.

Split	Examples	Accuracy	Grounded rate	Trajectory validity	Goal success	Artifact
test	500	1.0	1.0	1.0	1.0	results/20261008T005539Z_26822e7d16a44754af415c71c728034d
ood_test	500	1.0	1.0	1.0	1.0	results/20261008T005539Z_d7e71c7dd32e46e5864e15af0d53a552

These are deterministic reference infrastructure results, not learned model results.

No DER or neural data-efficiency evidence exists. Training time is null; no training occurred.

Not implemented

Neural training/downloads, learned baselines, experimental comparisons, interactive experiential learning, confidence intervals, UI, dream engine or published historical scores.

Known limitations

Rejection sampling is unbalanced and exhausts the finite 1x1 family in train; IID evaluation has only 2x1/2x2 examples. Full datasets/traces are in memory. Directory creation prevents overwrite but filesystem-write failures can leave partial directories. Future backends must implement identity/structure/validation/evaluation correctly. Stored schema checks are strict for core traces; benchmark metadata uses backend validation. Python model views are API isolation, not an adversarial filesystem sandbox. model_calls relies on model reporting; timeout records require the model to raise TimeoutError, not a runner-enforced wall-clock timeout. Training budget is metadata/subset identity only: future training must consume that subset. Ruff/MyPy/pytest and hosted CI were not run.

Relevant repository tree

```
configs/experiments/{iid,ood,data_efficiency}.toml
src/mindscape/core/serialization.py
src/mindscape/data/{schemas,backends,generation,splits}.py
src/mindscape/evaluation/{schemas,metrics,evaluator,runner}.py
src/mindscape/models/{base,reference}.py
scripts/{generate_data,dataset,evaluate}.py
datasets/generated/development_v1/{manifest.json,*.jsonl}
results/<unique_id>/{config.json,metrics.json,predictions.jsonl,summary.md}
docs/{benchmark,experimental_protocol,historical_references,milestone_02}.md
tests/{test_foundation,test_benchmark}.py
```

Git

History preserved on main. Phase-2 commits add infrastructure, verification tests, and documented development artifacts. Results remain available locally but are ignored by Git. Final working-tree status and latest commit checked after committing.

Milestone 03

PDF edition of preserved repository document | docs/milestone_03.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

First learned Mindscape milestone

IMPLEMENTED

Small trainable NumPy backend; answer-only baseline (A); trajectory-supervised state/history-conditioned policy (B); environment-interactive inference; action masking and unmasked diagnostics; resettable working memory; partitioned SQLite episodic API; prefix/final verification; deterministic Adam training, validation checkpoint selection; save/load and CLI; shared evaluator integration; tracking, failures and preliminary curves.

TESTED

All original 54 tests plus 15 learning tests passed. Coverage includes numerical gradient checks, optimizer learning/reproducibility, regime adapters, serialization, valid masks, memory reset, illegal actions, timeout/step limits, trajectories, verifier integration, configuration, checkpoint reload, SQLite partitions and train-to-evaluation smoke runs.

VERIFIED

10 actual trained checkpoints were reloaded before evaluation; 30 learned evaluations used identical held-out splits. Checkpoint/dataset/budget/regime compatibility is checked. The trained unmasked policy executes 19×3 to 57 with five legal actions and a complete verified trajectory. The baseline checkpoint predicts 28 for this separate demonstration. Each selected action, actual event/result and before/after state is retained in examples/19_times_3_learned.json. Working memory resets and no episodic retrieval is used.

EXPERIMENTS RUN

50 examples: baseline and policy at seeds 0,1,2. 100 and 250 examples: each at seed 0. Each model evaluated on 500 IID and 500 OOD examples; policy also evaluated unmasked. Two additional untrained unmasked controls (seed 0) reached 0% IID and 0% OOD success.

One early policy-training development smoke run is retained separately under results/.

No hyperparameters or test sets were changed after the fixed comparison started.

ACTUAL RESULTS

All values below are measured preliminary development results, not final thesis findings.

At 50 examples: mean ± sample standard deviation across three seeds; other sizes: one seed.

Training examples	Model	IID exact accuracy	OOD exact accuracy
50	baseline	0.267% ± 0.306%	0.000% ± 0.000%
50	mindscape	100.000% ± 0.000%	100.000% ± 0.000%
50	mindscape_unmasked	100.000% ± 0.000%	82.000% ± 4.084%
100	baseline	0.400%	0.000%
100	mindscape	100.000%	100.000%
100	mindscape_unmasked	100.000%	76.400%
250	baseline	0.400%	0.000%
250	mindscape	100.000%	100.000%
250	mindscape_unmasked	100.000%	78.800%

Policy supplied-trajectory validity is 100% for both masked and unmasked runs:
illegal proposals are rejected, so failed episodes retain valid incomplete prefixes.
Groundedness and goal success equal each policy run’s final accuracy; unsupported rate
is its complement. Baseline groundedness and goal success are 0%, unsupported rate 100%,
and trajectory validity null (no supporting trace). All five trained policies achieved
100% teacher-forced unmasked validation action accuracy. Seed-0 policy validation loss
at 50 examples fell from 1.409832 to 0.002929.

For the unmasked 50-example OOD policy: 106/67/97 invalid-action failures at seeds
0/1/2 respectively. Full taxonomy counts and every prediction are saved. No DER or
architecture-superiority conclusion is drawn. Masked 100% is guaranteed by one legal
action per state. The policy delegates exact arithmetic to the environment; the
baseline learns answer digits without that tool. Even unmasked action success does
not establish learned arithmetic or a fair data-efficiency advantage.

NOT IMPLEMENTED

Regime C feedback learning (interface only), pretrained language-model integration, PEFT/LoRA, memory-based learning/retrieval experiments, generated explanations, final statistical studies, matched-tool controls, multiple-choice environment redesign, UI, multimodality, dream engine and historical GPT/Gemini comparison.

KNOWN LIMITATIONS

Singleton masking and exact arithmetic access confound comparison. The baseline is very weak; failure is retained and cannot support general conclusions. More transition labels per policy problem and differing heads mean supervision/capacity are unmatched. Validation policy labels use correct histories (teacher forcing); OOD rollout errors expose different execution behavior. Development dataset is unbalanced and not a final untouched research set. Only 50-example points have three seeds. Training uses a fixed 400-update budget rather than equal epochs. The learned adapter is multiplication-only; core/benchmark remain general. Features support six-digit operands, eight-digit answers. TensorBoard and gradient-framework tooling are absent; plain JSON logs are used. Inference checkpoints do not preserve resume-training Adam state. Optional learning tests skip without NumPy. Floating point/RNG may differ across library/CPU versions. RAM could not be queried; PNG chart rendering is unavailable, so standard ReportLab SVG curves are delivered. Ruff/MyPy/pytest and hosted CI were not run.

MODEL USED

Local randomly initialized task-specific NumPy tanh MLP, not a foundation model. No downloaded model, API, pretrained weights or paid dependency. See docs/modeling.md.

MODEL SIZE

Shared backbone: 52 inputs, 64 hidden units, 3,392 parameters. Baseline: 8,722 total; policy: 3,652 total. Baseline eight decimal digit heads plus sign; policy four action logits.

DATASET SIZES

Development v1: train 1000, validation 250, IID test 500, OOD test 500.

Learning budgets 50/100/250 use identical recorded nested subset IDs for each model.

Policy row counts: 50 examples → 376 transitions, 100 examples → 740 transitions, 250 examples → 1783 transitions.

SEEDS

Dataset seed 42, fixed subsets. Training/mini-batch seeds 0,1,2 at budget 50;

seed 0 at budgets 100/250; untrained control seed 0.

TRAINING TIME

Ten fixed-comparison optimization runs total 0.553455 seconds.

Per-run seconds (optimization plus validation only):

Regime	Budget	Seed	Seconds
answer_only	50	0	0.076317
trajectory_supervised	50	0	0.035391
answer_only	50	1	0.074609
trajectory_supervised	50	1	0.035283
answer_only	50	2	0.074834
trajectory_supervised	50	2	0.035265
answer_only	100	0	0.077495
trajectory_supervised	100	0	0.034872
answer_only	250	0	0.074412
trajectory_supervised	250	0	0.034979

INFERENCE TIME

Thirty learned evaluations total 20.205644 seconds, including prediction/assessment.

Each run covers 500 examples. Full per-run times and calls remain in

experiments/first_learned/curves.json; model calls are 500 per baseline run and

3,944 IID / 6,635 OOD per complete policy run. Failed unmasked OOD runs use fewer calls.

TEST COUNT

69 passed on Python 3.12.14 / NumPy 2.3.5.

GIT STATUS

Branch main; prior history preserved. Final clean status checked after committing.

LATEST COMMIT

Training protocol committed before fitting: f33ce57; model implementation: ef3d94e.

This report and measured snapshots are included in the final milestone commit shown in chat.

REPOSITORY TREE

```
src/mindscape/models/{base,encoding,numpy_backend,learned}.py
src/mindscape/training/{adapters,optimizer,run}.py
src/mindscape/memory/{working,episodic}.py
src/mindscape/evaluation/{runner,evaluator,schemas}.py
configs/experiments/learning.toml
scripts/{train,evaluate,run_learned_episode,run_learning_experiment,plot_learning_curves}.
py
docs/{modeling,training,experimental_protocol,milestone_03}.md
examples/19_times_3_learned.json
experiments/first_learned/{curves,failure_analysis,fairness,untrained_controls}.json
experiments/first_learned/{test_accuracy,ood_test_accuracy}.svg
results/learned_development_v1/<model_budget_seed>/
  config.yaml, config.json, metadata.json, metrics.json, training.jsonl, checkpoint/
results/learned_development_v1/evaluations/<unique_id>/
  config.json, metrics.json, predictions.jsonl, summary.md
results/learned_development_v1/{comparison,curves,failure_analysis,19_times_3_learned}.jso
n
results/learned_development_v1/controls/<unique_id>/
tests/{test_foundation,test_benchmark,test_learning}.py
```

Milestone 04

PDF edition of preserved repository document | docs/milestone_04.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research-core milestone: controlled numerical-claim study

IMPLEMENTED

An additional numerical-claim environment and backend, preserving all previous phases. Identical-capacity answer-only/structured baselines and numerical-claim Mindscape B/C; non-singleton constrained/unconstrained execution, scalar-feedback experience collection and positive SQLite replay, model-visible/oracle separation, bounded tagged simulation, optional offline pretrained encoder/head backend, four-condition study orchestration, component interventions, robustness analysis, statistics, fairness audit and SVG plots.

TESTED

All prior tests retained. Added numerical-claim oracle cases, wrong-value non-correction, 100 candidates, target-mutation feature isolation, equal capacity, three-way evidence records, no Regime C target access, non-mutating dream rollouts, generic learned-transition adapter, action modes, open-loop evidence removal, serialization/distractor invariance, all-condition training/reload/shared evaluation, optional offline backend contract (mocked), final-attempt completion counters, statistical utilities and invalid configuration rejection.

VERIFIED

90 trained checkpoints; 204 common-runner evaluations: 168 primary and 36 component/robustness evaluations. Complete primary matrix: four conditions × seven budgets × three seeds × two splits. All predictions/configs/checkpoints/failures retained locally. 17 artifact-based integrity checks passed; primary candidate count is 101, constrained count is 100. No arithmetic correction channel or model-visible inference verifier judgment is present.

MODEL(S) USED

Local task-specific NumPy tanh MLPs. The prior model remains intact. Optional offline LocalHFEncoder/FrozenPretrainedBackend supports pretrained text features plus a trainable head and explicit download opt-in; no actual pretrained weights were trained/evaluated in this study because PyTorch/Transformers are absent. Its offline interface is tested with a mock encoder. No model download, paid API or foundation-model training occurred.

MODEL SIZES

All four primary conditions: 52 inputs, 64 tanh units, eight ten-class output heads plus sign; 8,722 trainable parameters each. Policy uses first two heads for proposed unit/carry and other heads as zero dummy labels. Exact capacity is matched, but usable label semantics/task difficulty differ. Same model family, initialization policy and optimizer.

TRAINING REGIMES

A: final-answer supervision. Structured baseline: static structured operands -> answer. B: local numerical claims supervised by correct training transitions. C: bounded exploratory proposals -> scalar accepted/rejected feedback -> replay of accepted own labels, with no reference answers/trajectories read. Fixed 600 Adam steps, batch 64, learning rate .003; validation is diagnostic only, not checkpoint selection. Online/no-replay intervention uses one gradient update per accepted proposal and is not a matched-compute memory ablation.

DATASET SIZES

Research v1 (new frozen dataset): train 1000, validation 100, IID test 200, OOD test 300. Nested budgets: 10,25,50,100,250,500,1000. Train structures: 1x1/2x1/2x2; OOD: 3x2/3x3/4x2. Canonical operand pairs are disjoint, including swapped pairs. Finite 1x1 family exhausts in train; actual per-family/carry distributions remain in difficulty reports.

SEEDS

Dataset seed 4242; training/experience seeds 0,1,2 for every primary budget/condition. All interventions use three seeds at budget 50. All conditions share exact subset IDs.

ACTUAL RESULTS

Observed performance is low. None of the requested accuracy/OOD/groundedness/goal or data-efficiency targets is met. The largest primary IID score of any single run is 8%.

The prior 82% exploratory OOD result is not reproduced under numerical-claim controls.

No architecture-only superiority or 2×/5× data-efficiency conclusion is established.

IID RESULTS

Mean ± sample standard deviation across three seeds, percentages:

Budget	Baseline A	Baseline B	Mindscape B	Mindscape C
10	0.000 ± 0.000	0.000 ± 0.000	0.667 ± 0.289	0.167 ± 0.289
25	0.333 ± 0.289	0.000 ± 0.000	2.333 ± 0.764	0.667 ± 0.289
50	0.500 ± 0.000	0.500 ± 0.000	3.333 ± 0.577	1.333 ± 0.577
100	1.000 ± 0.000	0.833 ± 0.289	4.000 ± 0.000	2.333 ± 0.764
250	0.333 ± 0.289	0.500 ± 0.000	4.333 ± 0.577	4.000 ± 1.000
500	0.333 ± 0.289	0.333 ± 0.289	4.000 ± 0.866	5.500 ± 2.291
1000	0.167 ± 0.289	0.333 ± 0.289	6.333 ± 1.443	6.000 ± 1.732

At budget 1000, Mindscape B groundedness/goal success = 6.167% ± 1.155%; C = 5.667% ± 1.443%. Answer baselines have no supporting trajectory: groundedness and goal success are 0%, supplied-trajectory validity is null. For policy conditions, full-trajectory validity equals grounded/goal rates in these runs. Some wrong claims cancel to a correct final answer, so exact accuracy can exceed groundedness.

Unsupported rate is the complement of groundedness. Every metric remains in aggregate.json.

OOD RESULTS

All primary conditions/budgets/seeds have 0% exact OOD accuracy and 0% grounded/goal success on the observed 300-example OOD set. This does not mean a universal probability of zero; individual-run Wilson intervals are stored in statistics.json. Per-structure and carry counts are reported in difficulty_failures.json. No held-out test tuning occurred.

DATA-EFFICIENCY RESULTS

N*(.80), N*(.90), N*(.95) and both baseline-to-Mindscape DERs are not_reached on IID and OOD. No interpolation or fabricated crossing. Ablation data-efficiency changes are not estimated from single-budget interventions.

ABLATION RESULTS

Mean IID accuracy changes at budget 50, percentage points (three seeds):

Intervention	Reference	ΔAccuracy	ΔGroundedness	ΔGoal success
constrained	trajectory	0.000	0.000	0.000
no_dream	trajectory	-0.333	-0.167	-0.167
no_environment_interaction	trajectory	0.000	-2.833	-2.833
no_episodic_replay	experiential	-1.333	-0.500	-0.500
no_state	trajectory	-3.333	-2.833	-2.833
serialization_distractor	trajectory	0.000	0.000	0.000

All OOD intervention changes are 0 percentage points. No-interaction preserves model-computed answers while removing actual evidence; the groundedness drop follows the metric definition, not proof of an arithmetic advantage. Serialization/distractor invariance is a deterministic feature-adapter property, not semantic distractor reasoning. Constant relation representation and mandatory external goal evaluator are not meaningful isolated ablations in this domain. B versus C varies trajectory supervision by construction. Reasons are recorded in ablations.json. The online/no-replay contrast also changes optimizer and update count; it cannot isolate memory causally at equal compute.

EXPERIENTIAL-LEARNING STATUS

Implemented limited batch off-policy self-generated feedback learning. At budget 1000, accepted replay labels are 738/874/1074 at seeds 0/1/2, versus 4506 supervised transitions per B run. Query budget: up to 64 per problem; exact counts are retained. No complete expert trace is delivered to C. Exploration is collected with initial weights, then positive feedback is replayed; no claim of online adaptive exploration or autonomous concept discovery. Sparse feedback and limited capacity are potential limitations, not

experimentally isolated explanations for poor performance.

DREAM-ENGINE STATUS

Runnable generic bounded branching with explicitly hypothetical states and deterministic or learned-callable TransitionModel adapters. Primary claim dreams use proposed-write dynamics, depth 2/branching 3, and learned confidence scoring without a numerical oracle. Goal-distance scoring is demonstrated separately in examples/dream_goal_demo.json: real state remains 0, goal is 3, selected first action is +1. A separately trained transition world model is not implemented. Simulated values never become observed evidence.

HISTORICAL-REFERENCE STATUS

Primary-source GPT-4/Gemini report records and exact prompting caveats are documented in docs/historical_references.md. GSM8K word problems are not comparable to our explicit integer multiplication task; historical checkpoint IDs are undisclosed and training budgets are not controlled domain sample counts. No victory claim or historical DER.

FAIRNESS CHECKS

17 computed checks passed: dataset integrity/disjointness, training IDs, equal parameter count, dataset hashes, balanced matrix, shared optimizer configuration, no target reads in C, no inference arithmetic-oracle calls, target-mutation isolation, >1 candidates, no inference correctness feedback, own-proposal actual values, hypothetical tags and train-only memory. This removes major earlier confounds, but hand-specified decomposition, different label/query counts, compute and trace accessibility remain explicit differences. No claim that architecture alone has been isolated. Exploratory paired McNemar tests at budget 1000, seed 0, after Holm correction: B vs A IID p=0.0001831, C vs A IID p=0.0390625; all other comparisons p=1. These within-seed tests concern small absolute IID differences versus a weak baseline and do not establish practical success or OOD benefit.

TRAINING TIME

32.965690 seconds across 90 trained primary/intervention models; actual per-run metadata retained.

INFERENCE TIME

114.325489 seconds across 204 evaluations; each covers 200 IID or 300 OOD examples. Includes model execution and evaluator assessment.

TEST COUNT

86 passed on Python 3.12.14 / NumPy 2.3.5; original tests preserved. Real optional PyTorch/Transformers integration, pytest/Ruff/MyPy and hosted CI were not run.

KNOWN LIMITATIONS

All primary OOD runs fail; IID success is far below targets. Task MLPs are small and not pretrained; optional pretrained interface was only mock-tested. The procedure/control schedule is hand-authored and narrows the scientific question to decomposed numerical learning. Goal/relation are fixed domain signals, not variable-goal or graph reasoning. Label/query budget and inference compute differ; replay uses only accepted labels and does not optimize rejected-action likelihood. Baselines cannot meet trace-groundedness without a trace emitter. Scalar C feedback uses a correctness oracle at training only, an intentional experimental condition. No final thesis conclusions or measured 2× DER.

A bookkeeping edge case undercounted completed C episodes in two original metadata files (1000/seed0: 23 vs stored 24; 1000/seed2: 76 vs stored 77). The audit retains both values; completion reporting here uses stored outcomes. Future code counts final-attempt completion correctly, with a regression test. Labels, weights and evaluation metrics were unaffected. Original experiment files were not replaced. Analysis-only chart drafts are retained; committed figures use corrected cost labels and uncertainty bars.
Local results occupy approximately 1.0 GiB, including replay/prediction detail. Checkpoints and full predictions stay local under ignored results/; measured summary snapshots are committed. Timing excludes dataset preparation and artifact I/O for evaluation; training includes experience collection/replay and fitting. RAM size was unavailable. PNG rendering is absent, so figures are standalone SVG vectors. Claims adapter is currently domain-specific.

GIT STATUS

Branch main; earlier code/history/results preserved. Final clean status checked after commit.

LATEST COMMIT

Protocol/run revision: df98891. Analysis/backend/diagnostics: 571147b. Final report and measured snapshots are committed separately; latest hash is reported in chat.

Diagnostic naming caveat: original C metadata used the generic key validation_metrics for training-replay diagnostics, not held-out validation. diagnostic_context.json sidecars identify the actual source without replacing original files; future metadata separates replay_metrics from validation_metrics. No C validation trace labels were used and no reported held-out benchmark metric depends on these training diagnostics.

REPOSITORY TREE

```
src/mindscape/environments/multiplication/claims.py
src/mindscape/data/claims_backend.py
src/mindscape/models/{study,study_encoding,pretrained}.py
src/mindscape/training/claims.py
src/mindscape/reasoning/simulator.py
configs/experiments/{research_dataset,research_study}.toml
datasets/generated/research_v1/{manifest.json,*.jsonl}
scripts/{train_research,evaluate_research,run_research_study}.py
scripts/{analyze_research_study,audit_research_study,run_dream_demo}.py
docs/{research_protocol,historical_references,milestone_04}.md
experiments/research_claims_v1/{aggregate,data_efficiency,ablations,statistics,...}.json
experiments/research_claims_v1/plots/*.svg
results/research_claims_v1/<condition_budget_seed>/{config.yaml,metadata.json,checkpoint,t
raining.jsonl}
results/research_claims_v1/evaluations/<id>/{config.yaml,metrics.json,predictions.jsonl,fa
ilures.json,summary.md,plots/}
examples/dream_goal_demo.json
tests/test_research.py
```

Modeling

PDF edition of preserved repository document | docs/modeling.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

First learned prototype

Hardware and model choice

Local inspection found an ARM64 CPU with 10 logical threads and NumPy 2.3.5.
The bundled Python runtime lacks PyTorch, Transformers, scikit-learn and TensorBoard.
A cached all-MiniLM-L6-v2 sentence encoder exists but is not an arithmetic generator
or action policy. RAM probing was denied by the sandbox; available RAM is unknown.
No GPU or download is required. The chosen model is a small task-specific NumPy MLP,
trained locally from seeded random initialization. It is not a foundation model or
language model. This is allowed by the discrete-classifier first-policy design.

Both models use the same 52-dimensional input space and 64-unit tanh hidden layer.
The answer-only baseline has eight ten-class digit heads (least-significant first,
zero padding) plus a two-class sign head: 8,722 trainable parameters. The trajectory
policy has four action logits: 3,652 parameters. The shared backbone has 3,392
parameters; output capacities differ and are recorded, not claimed identical.
Baseline feature slots for state/history remain zero. Neither encoder computes a
product. There are no stored lookup answers or pretrained proprietary dependencies.
Models support predict/save/load; the backend supports logits/generate/save/load.
Training is separate in training/, never in the benchmark or environment.

State-conditioned policy

Canonical version policy-state-1.0 includes problem, state, goal, valid_actions and
previous_action. Numeric features encode operand digits/signs/lengths, positions,
carry, partial row/total digits, length boundaries and previous action. Feature
selection deliberately excludes the phase string and legal-action identities from
network inputs: they directly reveal the next action in the current environment.

The serialized record still preserves those fields for action validation/masking.

Goal has a fixed domain meaning; a learned goal encoder is not implemented.

Four learned logits score multiply/flush/accumulate/finish. By default illegal choices are masked, then the chosen action is sent to the authoritative environment. The actual event/result/state updates working memory. Prefix replay verifies every transition, and final replay/independent multiplication oracle verify termination. No generated explanation is used. Step and wall-clock limits terminate failures. The explanation interface is secondary and not implemented in this milestone.

Critical identification limitation

The current environment exposes exactly one legal action in every nonterminal state. Masking therefore forces the correct procedural sequence, even with an untrained network. Masked 100% accuracy cannot establish learning, improved data efficiency, reasoning or generalization. The environment also supplies exact arithmetic unavailable to the answer-only baseline. This is an intentionally unmatched knowledge/tool condition, not a clean test of the research hypothesis.

We train the policy's unconstrained four-way cross entropy and report validation unmasked action accuracy. Additional unmasked episodes terminate on an illegal learned choice. These diagnostics exercise learned weights causally; an untrained masked control in tests confirms the confound. A meaningful future study needs multiple legal choices with consequences, matched tools/knowledge and proper controls. No superiority or DER conclusion is drawn from this prototype.

Memory and regimes

Working memory resets per episode and stores state, goal, latest observed feedback, actions, events, actual results and trajectory prefix. SQLite episodic memory supports store/retrieve/get/clear with explicit partitions; retrieval defaults to train and cannot see test entries unless specifically requested. Model inference does not use cross-episode retrieval or automatically persist evaluation answers. The API is an experimental foundation, not a demonstrated memory learning signal.

Regime A trains only problem operands -> answer digits/sign. Regime B trains each trajectory state -> next action, with prior executed action as history. It is trajectory-supervised and becomes environment-interactive at inference. Regime C has an explicit feedback-learning interface raising NotImplementedError; no claim of experiential or self-learning behavior is made.

Limits

Operand feature width is six decimal digits; answer width is eight. Larger inputs are rejected or unrepresentable. Training targets are range-checked; the included benchmark stays within those limits. Eight answer digits cannot cover every product of two six-digit operands. There is no neural arithmetic transition predictor, learned verifier, language understanding or transfer to a second domain. Plain NumPy checkpoints avoid pickle and reject nonfinite/wrong-shaped weights.

Controlled numerical-claim revision

The revised study is documented in research_protocol.md and milestone_04.md. Prior singleton/oracle-assisted results are exploratory, not current evidence. Primary numerical-claim execution writes model proposals without correction; Regime C uses training-only scalar feedback and positive replay. All four conditions share capacity and the evaluator; remaining procedural/supervision/compute differences are explicit. All reported thresholds are censored; no data-efficiency advantage has been established. Use the research-specific CLI/config files for this study.

Release Notes

PDF edition of preserved repository document | docs/release_notes.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Coding research completion

Exact scope: A100, structured100, B100, C100; 400 unique locked pairs; 1920 unique learning cases; 400 independently audited trajectories; 16 dedicated sequential latency cases. The 1122 original CPU rows and all retained checkpoints remain unchanged. Frozen batch16/float16/MPS evaluation resumed without the superseded operational deadline. Native inference and CPU/WASI tools are separately recorded.

Successful replay does not establish successful repair. Mixed precision/hardware, different supervision and interaction budgets prevent architecture-only causal attribution. Zero-budget, recovery and memory/dream component interventions were not measured. No GPT/Gemini superiority is supported. Read actual results and claims before interpreting completion as capability.

Reproducibility

PDF edition of preserved repository document | docs/reproducibility.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Reproduce preserved evidence

Use the pinned foundation snapshots and WASI runtime described in docs/coding/reproducibility.md. The emergency native entry point is scripts/coding/emergency_mps.py; its original deadline is fixed and it intentionally refuses late neural reruns. Saved per-episode files checkpoint completed keys; completed CPU episodes are never deleted. Explicit historical CPU reproduction uses MINDSCAPE_DEVICE=cpu MINDSCAPE_PRECISION=float32; accelerated evaluation requires MPS and rejects silent fallback. Filesystem, tokenizer serialization, test execution, checkpoint CPU backups and Git remain CPU utilities. Neural weights, forward/generation and LoRA tensors use MPS. Batch only independent requests across episodes; decisions within each episode remain sequential.

Recompute reports with work/final-venv/bin/python scripts/coding/emergency_report.py. Restore frozen evidence into an empty directory with scripts/coding/restore_completion.py --snapshot results/final/coding_research_v2 --destination PATH; use the actual snapshot name if the mandatory locked study remained partial. The restore utility verifies every manifest hash before copying. Foundation weights are external pinned downloads, not bundled; adapters, protocol, actual datasets, source, runtime and results are bundled.

The authorized full continuation is scripts/coding/complete_research.py; it removes the superseded operational cutoff, verifies the original protocol/source/checkpoint hashes, holds a single-runner kernel lock and resumes exact missing keys. See docs/research_continuation.md. The original emergency deadline and failure records are historical.

Research Conclusions

PDF edition of preserved repository document | docs/research_conclusions.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research conclusions

Hypothesis

Explicit observations, entities, relations, states, actions, events, results, goals, memory and verification might improve learning and structural generalization, especially problem-budget efficiency. Architecture existence is not evidence for this hypothesis.

Implementation

The final system adapts one frozen local SmolLM2 backbone under four conditions: conventional answer-only, structured answer-only, trajectory-supervised procedural B and binary-feedback experiential C. Procedural policies choose numerical claims among100 candidates. Real results execute those proposals unchanged, and the verifier judges after prediction. The multiplication cursor/decomposition is hand specified. Working/episodic/concept memory and bounded hypothetical simulation are implemented; conceptual interfaces are reusable, while current concrete schemas remain decimal specific.

Observed results

Exact per-budget/seed measurements, mean±sample SD, intervals, paired tests and failure groups are in docs/final_results.md and experiments/final_smollm2_v1/. The old negative phase-4 study remains a separate frozen historical result. The fresh final dataset and larger representation/head produce higher absolute procedural accuracy, but no final target or threshold-crossing efficiency ratio is assumed from that fact.

Interpretation

The experiments compare different information regimes, not only different wrappers. B gets several state/action labels per problem; C gets bounded scalar feedback on its own proposals; answer learners get one final-answer target. B's domain scaffold decomposes the numerical task. Any gain versus answer-only therefore cannot be attributed uniquely to Mindscape, memory, verification or dreams.

The matched-information conventional local supervised policy uses B's exact state/action training information and predictor while bypassing its memory wrapper. If its predictions match B, this does not establish an independent wrapper advantage. Groundedness is an evidence-output metric: absent trajectories cannot score as grounded, regardless of exact answer accuracy. State ablations establish only their measured condition/budget effects. Replay-storage and dream claims depend on their controls, not their existence.

Under the tested model, representation, supervision and compute budgets, the experiments do not establish a data-efficiency advantage or a broad structural-generalization advantage for the Mindscape architecture. Partial success at uncorrected local procedure execution is narrower than the proposed general hypothesis. The study is a rigorous measured outcome, including limitations and failed thresholds, rather than a claim that "Mindscape works."

Limitations

One arithmetic domain, one generated final dataset, three seeds, hand-defined sequencing, unequal label/query amounts, a frozen indexed-feature language backbone and cached CPU inference limit general conclusions. Statistical significance against an answer-only condition does not remove these information differences. A second learned vertical and full language-model fine-tuning were deferred. Historical GPT/Gemini GSM8K references are not direct comparators.

Future work

Use fully matched teacher information and interaction budgets across stronger sequence learners; test learned cursor/action planning; separate pretraining from categorical features and head capacity; design explicit carry-context holdouts before data generation; improve experiential exploration without providing full target trajectories; evaluate reuse rather than merely storage; benchmark cold compute; and add one small navigation backend with generic payload schemas. Any new work should pre-register its protocol and preserve these final artifacts.

Research Continuation

PDF edition of preserved repository document | docs/research_continuation.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research completion continuation

The completion request supersedes the emergency run's operational two-hour cutoff. The original frozen scientific protocol and historical emergency package remain unchanged.

complete_research.py resumes C first, then B's missing lockbox cases, remaining learning groups, and a dedicated 16-case latency study. It uses the already selected batch size 16 and float16 MPS path with one resident model/tokenizer. Dedicated latency measures sequential requests, whose actual batch size is 1, separately from batched evaluation throughput. Filesystem, tokenization, and WASI execution remain CPU operations.

A kernel-held file lock prevents simultaneous continuation launches. SIGINT/SIGTERM request a checkpointed stop after the active group; completed cases are fsynced individually. Restart the same command after an operational failure. Existing keys are skipped, scientific hashes are checked, and missing cases never receive invented scores. Failures are saved in results/coding/research_continuation_v1/failure.json; completion is recorded only after exact key-set checks.

Native execution is necessary: this tool execution environment cannot allocate MPS tensors, while the user's native Terminal has verified MPS availability and performed the earlier actual MPS evaluation. Native Terminal computer control was rejected for safety reasons. No CPU fallback is used.

Run from any native Terminal directory:

```
env -u PYTORCH_ENABLE_MPS_FALLBACK /Users/akilan/Documents/Codex/2026-10-08/files-pasted-
by-the-user-mindscape/outputs/mindscape/work/final-venv/bin/python
/Users/akilan/Documents/Codex/2026-10-08/files-pasted-by-the-user-
mindscape/outputs/mindscape/scripts/coding/complete_research.py >
/Users/akilan/Documents/Codex/2026-10-08/files-pasted-by-the-user-
mindscape/outputs/mindscape/work/research_completion.log 2>&1
```

After neural completion, independent replay, analysis, full tests, fairness checks, report reconciliation, immutable packaging, clean restoration, and hash verification are still required before the completion tag. The freeze gate requires 400 lockbox cases, 1920 learning cases, replay agreement, dedicated latency, and passing tests. A completion tag has not been created.

Verified prelaunch state: 296/400 lockbox cases (A100, structured100, B96, C0), 1122/1920 learning cases, unchanged frozen sources and byte-identical preserved CPU rows, no duplicate keys. Functional tests: 129 tests plus 6 subtests pass; scoped lint passes; fairness audit: 10158 assertions pass. Foundation pretraining exposure remains unknown; these checks do not establish universal contamination freedom.

Research Hypotheses

PDF edition of preserved repository document | docs/research_hypotheses.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research hypotheses

Explicit state-transition learning may improve data efficiency, OOD performance and evidence-supported output in bounded domains. This is untested. Deterministic algorithm performance alone cannot establish learned generalization or superiority.

Compare answer-only supervision (A), trajectory supervision (B), and concept plus limited examples plus environment feedback (C). C must not silently receive full trajectory supervision. Equalize backbones, test sets, examples, decoding, epochs and compute where practical; explicitly report additional procedural knowledge.

Research Protocol

PDF edition of preserved repository document | docs/research_protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Numerical-claim study protocol v1

Frozen config: configs/experiments/research_study.toml; frozen dataset: datasets/generated/research_v1. Both implementation and protocol were committed before held-out evaluation (df98891). Earlier results remain exploratory evidence.

Information audit

Condition	Model inputs	Labels/feedback at training	Inference feedback
Answer-only	Original operand digits/signs/lengths	Final answer digits/sign	None
Structured	Same operands plus static length/relation/goal representation	Final answer digits/sign	None
Mindscape B	Operand digits at current operation, previously self-proposed digit/carry, positions, goal/relation flags	Correct next numerical claim from training trajectories	Own written values/state only
Mindscape C	Same state features as B	Scalar acceptance/rejection of self-generated training proposals; accepted own values replayed	Own written values/state only

No model feature contains expected arithmetic values, target answer, future actual states, or corrected arithmetic. Primary execution writes the proposed number even when incorrect. The model never sees the numerical verifier judgment during inference. After an episode the common evaluator checks every claim and final goal privately. Regime C training has intentional scalar correctness feedback; its query count is logged. The verifier and target generator alone can evaluate the correct local arithmetic. Hypothetical future states are self-generated, explicitly tagged and never inserted into working/episodic observation memory as actual evidence.

All conditions use 52 inputs -> 64 tanh units -> eight ten-class heads plus sign, 8,722 parameters, same seeds, nested problem IDs, 600 Adam updates, batch 64 and

learning rate .003. Fixed final update is used for every major condition: validation labels/diagnostics do not select checkpoints. C receives no validation trace labels. No hyperparameter selection occurs after viewing IID/OOD results. The existing 82.0% exploratory OOD result is not transferred to this benchmark.

What this comparison can and cannot isolate

It removes corrected-arithmetic inference tools, singleton answer masks and capacity mismatches. The structured reduction to local multiplication/carry claims and automatic schedule are hand-specified procedural knowledge, not learned planning. A/B represent that knowledge differently: the intervention is representation/decomposition plus training signal, not proof that JSON or autonomous cognition causes an advantage. The concept seed describes decimal multiplication for every condition; task MLPs consume its fixed operation/goal encodings, not natural-language understanding. State features are functions of original observations and the learner's own previous claims. Supervision count and environment queries differ by training regime; sample budget comparisons must include those costs. Groundedness structurally requires a trace, so baselines lacking traces cannot establish it even if answers are correct. No memory or dream mechanism may consult held-out targets. Compute is recorded, not matched. These remaining differences must accompany every interpretation.

Action modes and numerical validation

Constrained: 100 syntactically legal numerical candidates (0..99), with no correctness mask. Unconstrained primary: those 100 candidates plus an invalid skip candidate. Every decision logs candidate count, proposed value, actual written result and a post-episode verifier judgment. A wrong arithmetic claim remains in the executed trajectory; no early correctness rejection rescues inference. Syntax-invalid actions terminate. Final verification checks both numerical rules and observation binding.

Regime C and replay

Exploration is bounded at 64 attempts per training problem. Each state is explored with

75% random untried candidates and 25% model scoring. Training rejection leaves the state unchanged, returns only false, and does not reveal the expected number. Success advances using the learner's own accepted number. SQLite records all state/action/event/result/next-state/reward attempts in the train partition. Positive-feedback replay supplies pseudo-labels equal to accepted learner proposals. No complete reference trajectory or reference answer is consumed. This is batch off-policy self-generated feedback learning, not general reinforcement learning or autonomous concept discovery. Model weights are updated after collection; exploration during that collection uses initial weights. A no-replay online variant updates each accepted experience once; its update count/optimizer differ and must be reported as a method contrast, not a clean matched-compute causal memory effect.

Simulation and ablations

Generic TransitionModel and tagged bounded DreamEngine accept deterministic or learned transition adapters. Primary numerical dreams branch three candidates at depth two; claim-write dynamics predict hypothetical states without a numerical oracle. They score model confidence, not ground-truth correctness. The standalone integer-navigation demo demonstrates explicit goal-distance scoring. Simulation is a deterministic prototype; a separately trained predictive world model is not implemented.

At budget 50, three seeds compare no state (retrained), no dream, constrained mode, no actual environment evidence (open-loop recurrence), serialization/distractor perturbation and online learning without episodic replay. Relation is constant in a single-operation domain, so graph attribution is not meaningful. External goal verification defines the common metrics and cannot be removed as an evaluator ablation. B versus C already varies trajectory supervision; it is not a single-component ablation. Single-budget ablations do not estimate data-efficiency changes.

Diagnostic naming caveat: original C metadata used the generic key validation_metrics for training-replay diagnostics, not held-out validation. diagnostic_context.json sidecars identify the actual source without replacing original files; future metadata separates replay_metrics from validation_metrics. No C validation trace labels were used and no

reported held-out benchmark metric depends on these training diagnostics.

Statistics and interpretation

Primary curves use three seeds at all seven budgets. Report means/sample standard deviations separately for IID/OOD and compatible conditions. Exact threshold crossing uses measured points only; censored N*/DER stays not_reached. Wilson intervals describe individual-run binomial accuracy, not cross-seed variance. Exploratory paired McNemar tests use one preregistered comparison budget (1000), seed 0, and Holm correction over six tests; do not pool repeated seeds as independent examples. Significance does not establish practical importance or isolate all remaining architectural differences.

The generated development research set is not an independently audited final thesis benchmark. Every failure and run directory is retained. No performance target is promised.

Research V2 / Cloud Workflow

PDF edition of preserved repository document | docs/research_v2/cloud_workflow.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

CPU cloud validation

Stage1 canonical release verification precedes this workflow. .github/workflows/tests.yml runs on GitHub-hosted Ubuntu24.04 with Python3.11.17, pinned action commit hashes, CPU PyTorch2.8.0+cpu and direct dependency pins. Install reports record resolved artifact URLs/hashes; linux-resolved.txt captures every installed version. The complete real Linux resolution is now pinned in configs/research_v2/linux-cpu-lock.txt. Run37939153417 installed successfully but failed18tests on an incompatible bundled ARM execution cache; failed evidence is preserved. The immutable WASM checksum is verified before preserving and rebuilding that cache for the actual host. Frozen source/data/outcomes are unchanged.

No benchmark inference, GPU allocation or paid reference API is invoked. LoRA unit tests use a tiny synthetic random model. The full export is stream-verified (all23842 hashes); only the immutable6242-file package is written and clean-restored, avoiding duplicated historical snapshots on limited disks.

```
python scripts/research_v2/restore_cpu_baseline.py --archive
research_checkpoint/research_release_2026-10-09 --destination work/cloud_research_v1
python scripts/research_v2/cpu_validate.py --evidence-root work/cloud_research_v1 --output
 results/research_v2/cloud_cpu_validation
# Repeat the identical command to validate resumability.
```

Each output namespace holds an exclusive lock. Code/config/rawdata/environment fingerprints reject changed inputs; completed steps are reused only after their output hashes match. Original experiment rows are never appended or rerun. Commands, logs, exit codes, host/backend/packages, resource use and hashes are uploaded as run artifacts (30day retention). Durable scientific receipts and the full Linux lock are versioned after independent retrieval; CI artifacts alone are not permanent publication storage.

Cloud success requires real Linux installation, complete export verification, all gates/tests/fairness/frontend checks, unchanged raw evidence and verified repeat invocation. Local success or active CI status is insufficient. Later inference/ablations/reference-model experiments need separately frozen protocols and outputs; historical CPU/MPS evidence remains unchanged.

Research V2 / Computation Tracing

PDF edition of preserved repository document | docs/research_v2/computation_tracing.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Actual computation tracing

The v2 diagnostic instruments the preserved learned numeric trajectory policy. It is a small trained NumPyMLP, not Qwen/GPT/Gemini inference. No coding-capability improvement is established by it. Its real checkpoint, configuration and input pairs are pinned in configs/research_v2/trace_diagnostic.json; checkpoint files are checksum-verified while streaming the complete preserved export.

Each JSON trace records exact model-facing input, numeric normalization and every feature vector passed to the real forward method; there is no text tokenizer in this backend. Actual logits, tensor/weight shapes, derived uncalibrated softmax, timings, deterministic working-memory changes, policy transitions, output, independent verification and execution failures remain explicit. The hidden activation summaries are recomputed from the observed input and checkpoint, not falsely described as stack-captured or causal explanations.

Structured entities/relationships, planning search, dreaming, feedback learning and recovery are not implemented in this numeric backend; the trace says so. Coding trajectories implement additional typed state fields but need separate prospective neural instrumentation. High masked success may be driven by deterministic singleton legal-action constraints. A trace cannot establish neural contribution or superiority: controlled interventions and independent scoring are required.

trace_diagnostic.py generates one machine-readable JSON and readable Markdown report per actual diagnostic execution. The observer-preservation test compares answer and trajectory against the original predictor; the intervention test confirms unchanged weights. No hidden target enters the model-facing view. Diagnostic pairs are not a held-out competitive benchmark. The next study must freeze novel evaluation tasks and all interventions before evaluation.

Research V2 / Gap Closure

PDF edition of preserved repository document | docs/research_v2/gap_closure.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Scientific gap closure: actual evidence and remaining limits

The original 1,920 numeric interventions, 734 failures and all 3,846 study hashes were reused. All 19,448 directly observed action selections had exactly one legal action. No completed intervention was rerun. Research v1's 400 locked cases, 1,920 coding learning rows and 400 saved audits remain unchanged.

Prospective controls and task-variant replication

Protocol/data commit 06d1f8e froze 24 novel identities, disjoint from all preserved numeric datasets and the previous 80-task v2 evaluation, before new inference. Three historical trained seeds executed four conditions: trained or zeroed copied weights, each with exact legal masking or permissive all-action selection. Original operands, environment, independent scorer and 256-action budget were fixed. No training, tuning, model download or external API was used. Cloud run 37949932056 (https://github.com/akilanpl/mindscape/actions/runs/37949932056), at 667585131e96967dc53037e4d0f50d94f55c020d, completed and safely resumed all 288 unique keys. Eighty-five unsuccessful cases remain in the evidence.

Novel family	Trained masked / permissive successes	Masking effect, percentage points	Task-cluster bootstrap 95% interval
Unsigned 3×2	24/24 versus 20/24	+16.7	0 to +41.7
Signed 4×2	24/24 versus 15/24	+37.5	+8.3 to +66.7
Unsigned 1×4	24/24 versus 24/24	0	0 to 0

Task clusters, rather than 24 executions per family, are the resampling units: eight distinct tasks per family, conditional on three shared historical checkpoints. All 18 predeclared per-seed tests were Holm-corrected; none passes 0.05 (minimum adjusted p=0.140625). The small replication does not establish confirmatory significance. Degenerate bootstrap intervals at observed ceilings do not establish population equivalence.

Zero-weight policies succeeded with masking on every task and failed under permissive selection on every task. Seed copies have identical zero weights and are not independent controls: the effective comparison is 24/24 unique masked tasks versus 0/24 permissive tasks. The observed paired effect is +100 percentage points. An explicitly post hoc, conservative 95% interval is +66.6 to +100 points, using simultaneous Bonferroni Clopper-Pearson bounds on beneficial and harmful paired outcomes. Its sampling interpretation is confined to these three numeric task families, not arbitrary task distributions. This descriptive supplement does not replace the predeclared tests or their multiplicity correction.

The zero-weight intervention supports a causal statement confined to this implementation and these tasks: trained neural weights were unnecessary for goal attainment when the controller's exact legal-action filter remained enabled. Disabling that filter changes action selection while preserving the same environment/scorer. It exposes dependence on learned-policy behavior; it is not a harder scorer or a new broad-intelligence benchmark. The qualitative masking finding replicated across newly frozen task variants. It did not replicate a uniform learned-policy gap: the 1×4 family showed none. Models, implementation and training pipeline are shared, so this is task-variant replication, not independent laboratory or cross-domain replication.

Coding-foundation computation telemetry

A hash-bound, nonduplicating index covers all 2,320 saved coding episodes. It links exact raw rows, model calls, responses, timings, failures, component actions, state transitions, adapters and protocol references. Historical exact serialized prompts, token IDs and logits were not retained; they remain unavailable rather than retroactively fabricated.

Two successful observer diagnostics used the existing pinned Qwen2.5-Coder-1.5B weights and original budget-100 adapters, on CPU float32 without generation caching. The same historical lockbox task was reused for instrumentation, not added to accuracy evidence:

- Model-only: one model call, 39 direct token-input/logit-output pairs; repair unsuccessful due to an invalid edit envelope; elapsed including loading 9.25 seconds, process peak RSS 8.46 GB.

- C condition: eight model calls, 352 direct token/logit pairs; eight retrieval, dreaming and planner invocations, three actual edits and three memory updates. Final hidden-test evaluation passed. Five later proposals hit the original edit budget. Elapsed including loading 71.98 seconds, process peak RSS 7.16 GB.

These are observed process memory maxima, not aggregate system memory. Timings include instrumentation and differ from original MPS benchmarking; no speedup or accuracy estimate is inferred. The first diagnostic failed because PEFT generation bypassed an outer hook. Its failure log is retained; the corrected observer hooks the actual base model. No tensors or lost response were invented for that failed diagnostic.

Live traces close exact-input, direct tokenization/model-forward, selected logits/probabilities, component invocation, state-transition, output, timing and failure-observation gaps for the tested paths. Probabilities are derived and uncalibrated; dreaming/planner predictions remain hypotheses. Hidden activations and live coverage of every historical adapter/condition are not measured. Historical input/logit gaps cannot be closed from saved rows alone. The reusable observer and report renderer are separate from unchanged v1 source. Observer reporting and exception handling evolved after the diagnostics; stored events and model/checkpoint hashes remain immutable.

Validation and reproduction

Cloud validation passed 138 tests plus six subtests, with zero failures, errors or skips. Saved v1 integrity, fairness, frontend playback, scoped lint and safe resume passed. Two focused local observer tests pass, including PEFT base-forward capture, absence of extra forwards, memory observation and cleanup after an exception. Independent verification checks all new trace/outcome hashes, all historical pointers, the original study manifest, direct legal-action cardinalities, raw/statistic agreement and exact paired tests by independently enumerating every fair binary allocation of the observed discordant pairs.

The immutable gap-closure export stores the cloud artifacts, both actual coding traces, readable reports, environments and failed diagnostic log. Cached model/adapter/tokenizer/teacher hashes were independently rechecked unchanged. Existing v1 and v2 archives remain preserved. Restore using scripts/research_v2/evidence_archive.py --verify-existing; the release manifest records the concrete archive path and checksum. Then:

```
python scripts/research_v2/verify_gap_closure.py --package work/gap_closure_restore --old-
study work/numeric_study_restore/mechanism_study --historical-evidence
work/cloud_research_v1 --receipt work/gap_closure_verification.json
python scripts/research_v2/coding_telemetry.py --trace
work/gap_closure_restore/coding_probe_c/trace.json --report work/coding_trace_report.md
```

The historical evidence and numeric study are restored by the already validated v1 and numeric archive utilities. Verification performs no neural inference. Live diagnostics require existing checksum-pinned local weights; absent weights are an explicit dependency, not permission for a download.

This phase establishes bounded mechanism evidence and targeted coding observability. Cross-domain independent replication, all-adapter live coverage and historical GPT/Gemini comparison remain unrun. Exact reference releases, access and an authorized API budget are still unspecified. No Research v2 completion tag or general-model superiority claim is justified. No public website was built.

Research V2 / Mechanism Study

PDF edition of preserved repository document | docs/research_v2/mechanism_study.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Frozen numeric mechanism study

This study tests the existing 52-input, 64-hidden-unit NumPy trajectory policy at fixed trained weights. It does not evaluate the coding foundation model, establish general intelligence, or compare GPT/Gemini.

The protocol and novel evaluation data were committed before execution (e059e8e). Eighty task identities are disjoint from all preserved generated dataset identities and each checkpoint's training IDs: 40 unsigned 2×2 IID tasks and 40 signed 4×3 OOD tasks. This identity check cannot establish absence of every possible semantic overlap or outside pretraining exposure.

Three historical budget-50 checkpoints (seeds 0, 1, 2) each execute all eight conditions of legal-action masking, state-feature readout, and previous-action-feature readout. Removing a readout means zeroing its declared input slice, not removing the entire memory architecture. Operands, model weights, tasks, environment, independent scorer and 256-action budget stay fixed. Verification and goals are controls; planning, dreaming and recovery are absent from this backend.

Each of the 1,920 planned executions stores actual neural inputs, logits, derived uncalibrated probabilities, direct action-selection observations, memory changes, result and independent verification. Hidden activations are reconstructed and labelled; they are not causal explanations. Executed failures remain unsuccessful outcomes. Missing executions never become fabricated zeros. Instrumented runtime includes observation overhead and is not a dedicated latency measurement.

Three primary comparisons use the all-enabled condition against each single feature removal. Exact paired McNemar tests are calculated per seed and split with Holm correction across 18 predeclared tests. Task-cluster bootstrap confidence intervals concern task sampling conditional on these three trained seeds. Wilson intervals accompany per-cell rates. These intervals do not imply generalization to arbitrary domains or model seeds.

Cloud execution commands (after checksum-verified archive restoration):

```
PYTHONPATH=src python scripts/research_v2/mechanism_study.py --evidence-root
work/cloud_research_v1 --output results/research_v2/cloud_cpu_validation/mechanism_study
python scripts/research_v2/analyze_mechanisms.py --study
results/research_v2/cloud_cpu_validation/mechanism_study
```

Run identity binds source, protocol, tasks, checkpoint hashes and runtime. An exclusive lock prevents concurrent writers. Every case is saved before its fsynced canonical row; same-identity orphan traces can be recovered. Repeating the study verifies completed trace checksums and skips completed keys. Different identity in the same namespace is rejected.

Results must be interpreted within this explicit software intervention design. Ordinary-test/OOD differences, failed cases and any weak or null mechanism effects must be retained. Broad foundation-model telemetry and historical-model comparisons remain outstanding. Exact GPT/Gemini releases, access and authorized spending are required before those comparisons can run.

The actual run and findings are in the scientific report (scientific_report.md). Restore every saved cloud trace and result from a fresh clone without inference:

```
python scripts/research_v2/evidence_archive.py --verify-existing --archive
research_checkpoint/research_v2_numeric_2026-10-09 --restore work/numeric_study_restore
```

The checksum-bound dashboard is mechanism_study/dashboard.html inside the restored evidence. The same report and raw statistics are versioned under experiments/research_v2/cloud_runs/37946027443/mechanism_study.

Research V2 / Reproduction

PDF edition of preserved repository document | docs/research_v2/reproduction.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research v2 reproduction baseline

Canonical public repository: https://github.com/akilanpl/mindscape. Default branch mindscape-cloud-migration-2026-10-08; frozen Researchv1 tag mindscape-research-complete-v1 points to 2862f5ae54443b231abda4255bdd6c652eca2343. Earlier releases and failed studies remain preserved.

Clone, then restore the checksum-bound release into an empty destination:

```
git clone https://github.com/akilanpl/mindscape.git
cd mindscape
python scripts/coding/migration_checkpoint.py restore --root
research_checkpoint/research_release_2026-10-09 --destination work/research_v1_restore
```

Source/tests/protocols/reports and archiveindexes are tracked. Large evidence/adapters live in immutable Git-versioned32MiB parts, with SHA256 in ARCHIVE.json and SNAPSHOT_MANIFEST.json. Foundation weights are pinned external HuggingFace revisions in the v1 manifest and bootstrap, not embedded credentials or copied virtual environments.

Stage1 clean Git clone and complete archive restoration passed 130tests plus6subtests in the existing pinned Mac environment. This does not establish a clean Linux dependency installation: that is Stage2. The old unittest-only CI failed on missing numpy/pytest and is preserved in Git history. See experiments/research_v2/stage1_canonical_repository.json for exact scope.

Research V2 / Scientific Report

PDF edition of preserved repository document | docs/research_v2/scientific_report.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research v2: verified progress and bounded mechanism findings

Research v2 is not fully complete. The subsequent scientific gap-closure report (gap_closure.md) records targeted replication and real coding-foundation telemetry; the original findings below are preserved. Git preservation, clean v1 restoration, actual Linux cloud validation, numeric-core traces, prospective task freezing and a controlled numeric intervention study are verified. Coding-foundation telemetry, broader domain replication and exact historical GPT/Gemini comparisons remain outstanding.

Provenance and validation

Research v1 remains at 2862f5ae54443b231abda4255bdd6c652eca2343 with its original completion tag and immutable archives. No frozen implementation under src was changed. The canonical restoration verified all 23,842 archived file hashes. An initial Linux run failed because an archived Apple ARM compiled WASI cache was incompatible; that failed evidence is retained. Recompilation from the same checksum-pinned WASM resolved portability without altering the frozen package.

Cloud run 37946027443 (https://github.com/akilanpl/mindscape/actions/runs/37946027443) executed commit 0c3f869077756f50e89a551a2fa2b9ca58f05728 on Linux x86_64 with four CPUs and Python 3.11.17. Dependency pins include CPU PyTorch 2.8.0+cpu; the actual studied NumPy MLP uses float64 CPU arithmetic. The full suite passed 137 tests plus six subtests, with no failures, errors or skips. Saved-evidence integrity, statistics, fairness, frontend playback, scoped lint and resume checks passed. Separate study peak memory and compute cost were not measured; validation-process memory is not a substitute.

Frozen experiment and actual outcomes

The protocol and 80 novel, identity-disjoint tasks were committed before execution. Forty IID unsigned 2×2 and forty OOD signed 4×3 multiplication tasks were evaluated for three preserved budget-50 trained seeds across eight binary conditions. These are 1,920 repeated task/seed/condition executions, not 1,920 independent tasks. Every completed key was unique. All checkpoint hashes remained unchanged. Repeating the study verified existing traces and did not repeat inference.

Mask/state/previous-action condition	IID successes / 120	OOD successes / 120
000	0	0
001	0	0
010	61	2

Mask/state/previous-action condition	IID successes / 120	OOD successes / 120
011	120	43
100	120	120
101	120	120
110	120	120
111	120	120

All 734 unsuccessful executions are retained with actual invalid-action outcomes. No unrun case was scored as zero.

The masked controller achieved all goals even when both neural feature readouts were removed. The deterministic environment performs arithmetic and the legal-action mask can identify the next valid operation. Consequently, masked success does not demonstrate learned arithmetic or neural necessity. With both readouts present and masking removed, IID remained 120/120 while OOD fell to 43/120 (35.83%). The observed OOD gap under this controlled intervention is 64.17 percentage points; the task-cluster bootstrap 95% interval is 50.0-77.5 points, conditional on the three trained seeds. Per-seed exact paired McNemar tests use a predeclared Holm family of 18 comparisons; full values and cell Wilson intervals are in the machine-readable statistics.

With masking enabled, the two single-readout primary interventions had no observed effect. Their empirical bootstrap intervals collapse to zero because all paired observed differences are zero; this is not proof of universal equivalence. The unmasked condition table shows feature interactions descriptively: neither previous-action features alone nor the operand-only readout produced a successful case. No unadjusted significance claim is made for exploratory interactions.

Observation and interpretation

Each execution has JSON and readable Markdown with exact numeric input, declared normalization, model/checkpoint identity, feature slices, neural logits, derived uncalibrated probabilities, shapes, actual action selection, working-memory changes, trajectory and independent result. No text tokenizer is present. Hidden activation summaries are explicitly reconstructed from observed features and checkpoint weights rather than falsely stack-captured. They do not explain causal influence. Instrumented episode time summed to 37.53 seconds; it excludes serialization and setup, includes observer overhead, and is not a dedicated latency benchmark or hardware speedup claim.

Feature-readout removal is not removal of the entire memory module. Verification and goals were held fixed. Planning, dreaming, feedback learning and recovery are absent from this backend. The experiment establishes effects of declared software interventions at fixed trained policies in this numeric environment, not broad cognitive or coding superiority. Training/evaluation identity separation was verified; semantic overlap and external pretraining exposure cannot be excluded universally. Historical unsuccessful numeric and coding evidence remains preserved.

Durable artifacts and outstanding gates

The versioned archive research_checkpoint/research_v2_numeric_2026-10-09 contains all cloud receipts and per-input evidence: 3,876 files, 11,957,855 compressed bytes, independently hash-verified after clean restoration. Its manifest binds every file and the exact source commit. Raw indexes, statistics, dashboard,

tests and job receipts are under experiments/research_v2/cloud_runs/37946027443. Stage receipts explicitly identify the verified scope.

Exact historical GPT/Gemini releases, access conditions and an authorized spending limit have not been supplied. Their outcomes, latency, costs and performance gaps are unknown, not zero. The coding-foundation model and its 61 adapters remain in preserved v1 evidence, but their deep computation telemetry and controlled v2 interventions have not run. Independent broader-domain replication remains necessary before claiming scientifically competitive architecture results. No full Research v2 completion tag is created.

Subsequent scoped scientific closure

See actual controls, coding telemetry and remaining limitations (gap_closure.md). Original numeric evidence was reused; 288 additional prospectively frozen control executions and two observed coding diagnostics were added. No full v2 completion claim is made.

Second Vertical

PDF edition of preserved repository document | docs/second_vertical.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Second vertical status

A second learned environment is deferred. The final priority is complete multiplication validation, fair controls and frozen evidence. Generic Environment is a structural, type-parameterized protocol and the runner delegates domain-specific target verification to dataset backends. Both implemented multiplication environments satisfy the interface. The bounded integer-position goal demo reuses the simulator without arithmetic-model changes and explicitly tags hypothetical states; it is not a second-domain learning/transfer result.

A future grid-navigation backend can reuse Entity, Relation, Goal, Event, evidence kinds, the environment protocol, dataset identity/leakage machinery, runner/metrics, memory storage and dream engine. It will need an appropriate state/observation/action payload and domain verifier, because current concrete decimal schemas include multiplication fields. The final claims table therefore does not claim demonstrated architecture transfer.

Training

PDF edition of preserved repository document | docs/training.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Local reproducible learning

Install optional .[learning] dependencies (NumPy); .[plots] adds ReportLab for

standalone SVG charts. Existing deterministic core remains dependency-free.

```
export PYTHONPATH=src
export OPENBLAS_NUM_THREADS=1
export VECLIB_MAXIMUM_THREADS=1
python scripts/train.py --dataset datasets/generated/development_v1 \
  --config configs/experiments/learning.toml --kind baseline --budget 50 --seed 0 \
  --output results/my_baseline
python scripts/train.py --dataset datasets/generated/development_v1 \
  --config configs/experiments/learning.toml --kind mindscape --budget 50 --seed 0 \
  --output results/my_policy
python scripts/evaluate.py --dataset datasets/generated/development_v1 \
  --checkpoint results/my_baseline/checkpoint --regime answer_only --split test
python scripts/evaluate.py --dataset datasets/generated/development_v1 \
  --checkpoint results/my_policy/checkpoint --regime trajectory_supervised --split
ood_test
python scripts/evaluate.py --dataset datasets/generated/development_v1 \
  --checkpoint results/my_policy/checkpoint --regime trajectory_supervised --split test
--unmasked
python scripts/run_learning_experiment.py --dataset datasets/generated/development_v1 \
  --output results/my_comparison
python scripts/plot_learning_curves.py results/my_comparison
```

Output paths must be new. Evaluation verifies checkpoint dataset hash, training

budget and regime against the same dataset manifest. Training reads only the recorded

nested training subset and validation labels. The held-out sets are validated for

integrity/leakage but never passed to fitting or checkpoint selection. Seeds change

weight initialization and mini-batch sampling; the dataset/subset IDs remain fixed.

Fixed preliminary config: 400 Adam updates, batch 64, learning rate .003, validation

loss checks every 50 updates. Choose the minimum validation-loss checkpoint. Both

systems use identical optimizer steps, batch size, seed policy and training problem

budgets; the policy receives multiple supervised transition labels per problem.

The minibatch units therefore differ; label counts, selected step and runtime are

recorded. No hyperparameter tuning on IID/OOD scores occurs in the comparison script.

Training outputs: config.yaml (JSON syntax, valid YAML 1.2) and matching config.json with complete dataset manifest, metadata.json,

metrics.json, training.jsonl and checkpoint/{weights.npz,backend.json,model.json}.

Evaluation uses the existing runner and saves full configs, metrics and per-example predictions with action/transition diagnostics. Comparison outputs retain fairness conditions, raw curves, failure counts and the full 19×3 learned demonstration.
All run directories refuse overwrite. Reported training time measures optimization and validation only, excluding data loading/adaptation/checkpoint I/O. Existing inference time includes model execution and per-example assessment, not loading or artifact writes. Model calls count one per baseline example or per policy decision.

Checkpoint reload is tested and used before benchmark evaluation. Resume-training optimizer state is not saved; restarting a training run uses the recorded config/seed rather than continuing Adam moments. Reproduction is exact on the tested NumPy/CPU setup for weight arrays and predictions. Different NumPy/BLAS/CPU versions can cause floating-point or RNG differences; versions and Git revision are recorded. Timing, UUIDs and timestamps naturally differ. NumPy's random Generator has no cross-version compatibility guarantee: https://numpy.org/doc/stable/reference/random/generator.html

TensorBoard is unavailable; plain machine-readable logs provide tracking. No external tracking service or paid API is used. The SVG plots are labelled preliminary and retain raw data; 50-example points average three seeds, larger points use one seed. Final statistical studies and confidence intervals remain future work.

Controlled numerical-claim revision

The revised study is documented in research_protocol.md and milestone_04.md.
Prior singleton/oracle-assisted results are exploratory, not current evidence.
Primary numerical-claim execution writes model proposals without correction;
Regime C uses training-only scalar feedback and positive replay. All four conditions share capacity and the evaluator; remaining procedural/supervision/compute differences are explicit. All reported thresholds are censored; no data-efficiency advantage has been established. Use the research-specific CLI/config files for this study.